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1 Abstract

As part of the First-year Electronics Design Project, our team has been tasked to design, build, and test a remotely controlled rover equipped with sensors and a user interface for real-time control and data communication. The objective was to build a working prototype that was lightweight, robust, reliable, cost-effective, and user-friendly to survey and identify ducks within a designated area.

The rover is integrated with four main sensors – radio, magnetic field, infrared, and ultrasonic sensors. These sensors and their designed circuits are then connected to a microcontroller, which processes the data received from the ducks. The data is then transmitted wirelessly to an app UI, developed with JavaScript, to present the relevant information to the user. The app is also capable of controlling the rover's movements in all directions.

This project demonstrates effective integration of hardware and software components and showcases what our team has learned this year with regards to circuit design, prototyping and testing, as well as programming and digital data representation. The result is a rover that can obtain the relevant information of each duck, manoeuvre quickly and accurately via app control, as well as relay information back to the user.

2 Introduction

2.1 Outline

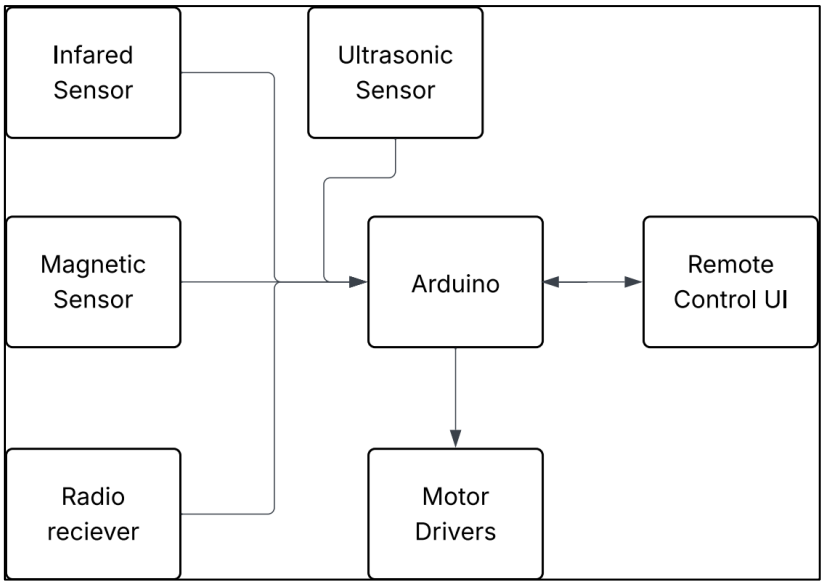


Fig 2.1: Overarching System Flowchart

In this project, the task is to design, build, and test a rover that is remotely controlled to approach and obtain information about the ducks in a play area. To do so, four different sensors and their respective circuits need to be built to detect the information. These sensors would need to be able to detect radio waves, magnetic field direction, infrared waves, and ultrasonic waves. The detected information would then need to be sent to a microcontroller, thus requiring software and code to process it to send it to the user. The rover needs to be mobile as well, thus requiring both software and hardware to control its movement on the play area.

2.2 Project Management and Planning

Based on the above specifications and requirements, we split our group equally into two main teams - a sensors team and a mechanical and UI team.

The sensors team would focus on researching and creating or purchasing the required sensors. The team will then proceed to design the relevant circuits and ensure the simulation results achieve the required results. Afterwards, the team will build the full sensor setup and perform rigorous testing on it.

The mechanical and UI team will research the Adafruit Metro M0 Express board to understand how to utilise it to meet task requirements. The team's first task was to fix up the EEEBug for initial tests and evaluation. From there, the team will write code to enable our app UI to control the direction and speed of the rover, as well as decode the detected inputs to send to the user.

At the same time, the team will design and manufacture new chassis and mechanical components required to accommodate the sensors and circuits, keeping in mind to satisfy the size and weight requirements as well.

The teams will then come together to integrate all systems and perform final tests.

The proposed timeline to meet all task requirements is as shown below in Fig 2.2. The timeline took into account research, trial and error stages, integration, and testing. It also included sufficient buffer time for any issues and debugging.

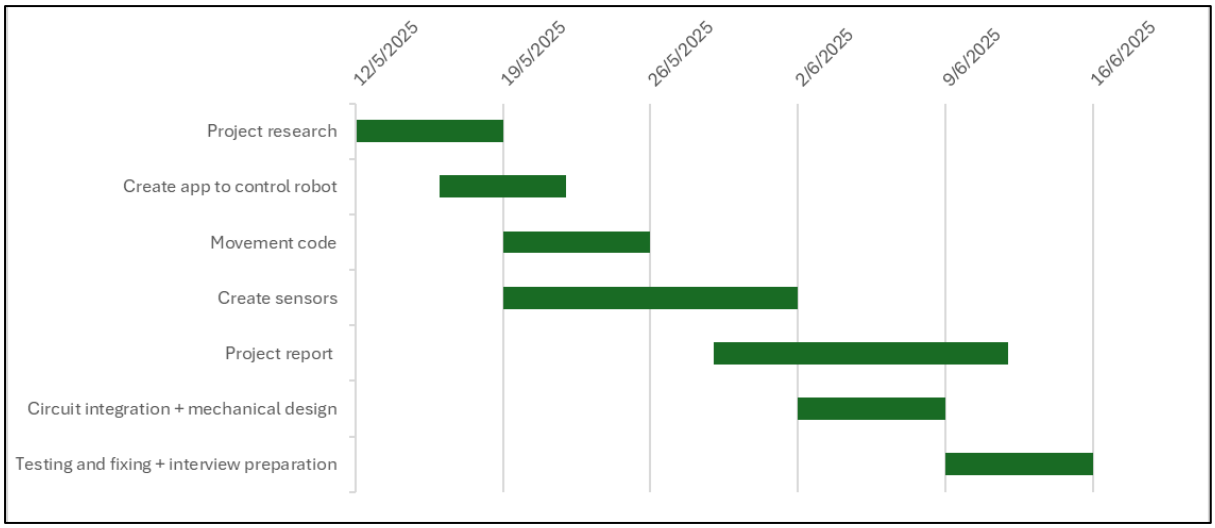


Fig 2.2: Gantt Chart for project timeline

The teams managed to follow the proposed timeline above and completed all tasks.

2.3 Design Criteria

For each of the sensors, the detected data may likely be too weak for direct processing. Hence, amplification would be required before demodulation and other processing. The amplification would also reduce high requirements and workload for the mechanical team, as there will be fewer restrictions on how the rover needs to approach and receive information from the duck. This will increase the sensitivity of our sensors so that the microcontroller is able to interpret the demodulated signals more accurately and efficiently. Thus, the sensor team needed to research and purchase an op-amp with suitable specifications, such as the gain bandwidth product.

Each sensor would require filtering and processing as well, which is discussed in its respective sections in the report.

Fig 2.3 is a table for the species classification of the ducks.

Species	Infrared (Hz)	Radio (Hz)	Magnetic Field Direction
Wibbo	457		Down
Gribbit		100	Down
Snorkle	293		Up
Zapple		150	Up

Fig 2.3: Duck Species Classification Table

The data received from each duck will determine its species, which needs to be identified. Hence, this determined the general flow and concept of the code for the microcontroller's data processing. This is further discussed under Section 8 - Remote Control UI.

3 Radio Sensor

3.1 Design Overview

In this subsystem, the task is to detect an Amplitude Modulated (AM) radio signal, demodulate the signal, which will then be fed into the microcontroller for frequency detection.

To execute this, an inductive antenna is built as part of a resonant RLC circuit, which will resonate at the carrier frequency of the AM radio signal. The output of this signal will then have to be amplified before the signal is demodulated. The demodulation involves rectification of the amplified signal before passing it through a low-pass filter.

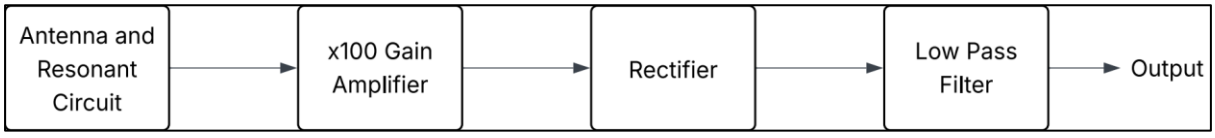


Fig 3.1: Radio Sensor System Flowchart

The radio waves emitted by the duck are an AM radio signal, consisting of the carrier and message signals. A coiled antenna needs to be built out of wire to form a resonant RLC circuit, which oscillates in the presence of radio waves of the correct frequency, which in this case is the frequency of the carrier signal, 89kHz.

The signal amplitude generated by the coiled antenna is likely to be small and hence requires amplification before processing the data. The microcontroller is also unable to sample the radio signal at its carrier frequency, and hence, rectification and demodulation are required to reproduce the message signal. However, due to the voltage drop during rectification, the

amplification needs to be high enough for demodulation to occur. With the above data, the sensors team designed the following circuit in Fig 3.2.

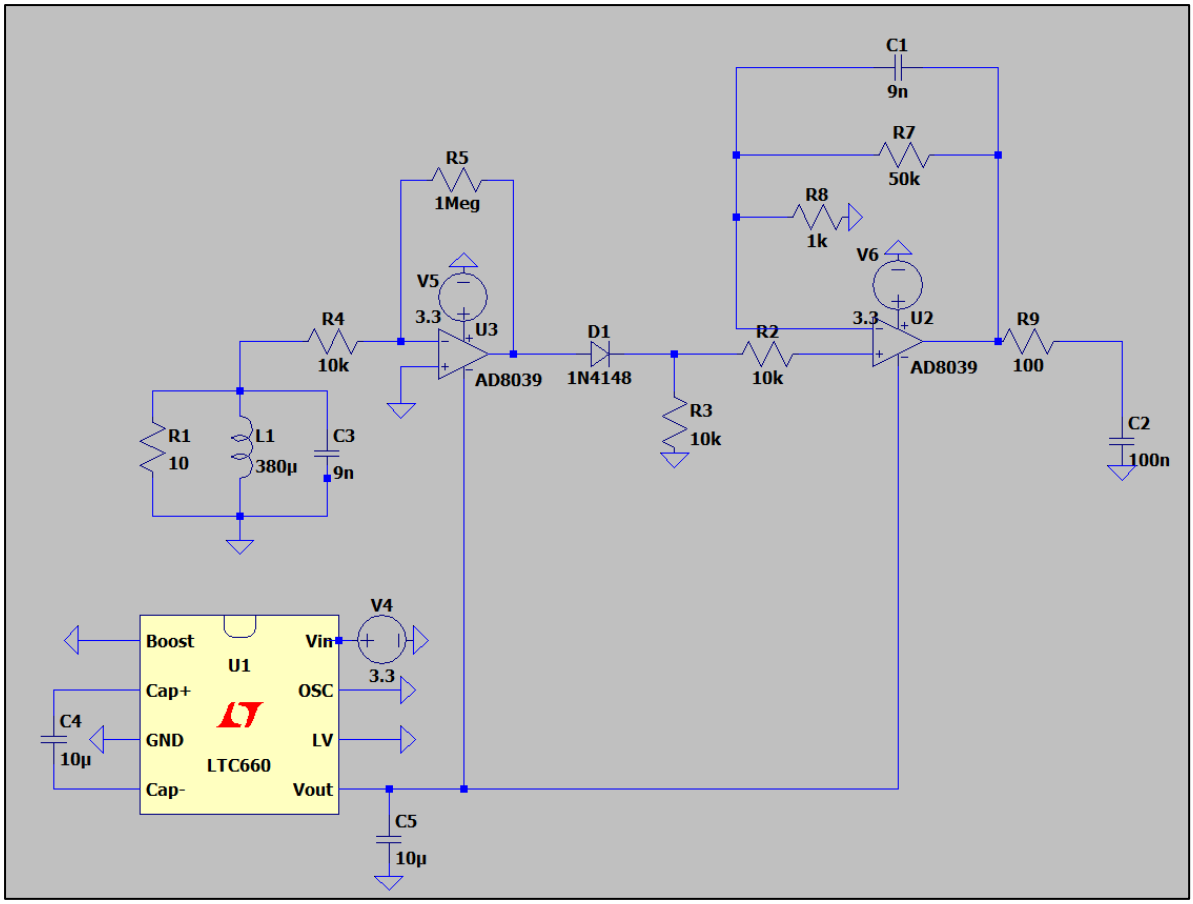


Fig 3.2: LTspice Diagram for Radio Sensor Circuit

The RLC values are determined based on the inductance of the coiled antenna, measured by the LCR bridge, and this is discussed further in the next subsection. The output is then amplified using an inverting amplifier with a gain of 100, which is then rectified and demodulated. The final output after demodulation is then sent to the microcontroller as an analogue input for processing.

After simulating the circuit on LTspice, the following output in Fig 3.3 is obtained.

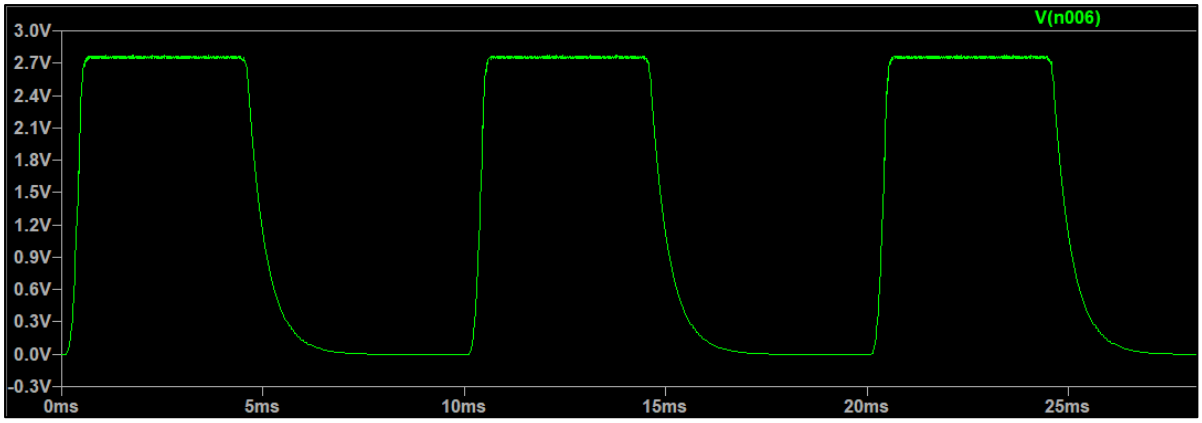


Fig 3.3: LTspice Simulation Output for Radio Sensor Circuit

3.2 Technical Solutions and Build

An LCR circuit was implemented at the input stage, where the inductor (L) was represented by a coil. The capacitance (C) value was then calculated based on the target carrier frequency of 89 kHz as shown below to ensure proper resonance.

$$\omega_c = \frac{1}{\sqrt{LC}} = 89000$$

$$LC = 3.1979 \times 10^{-12}$$

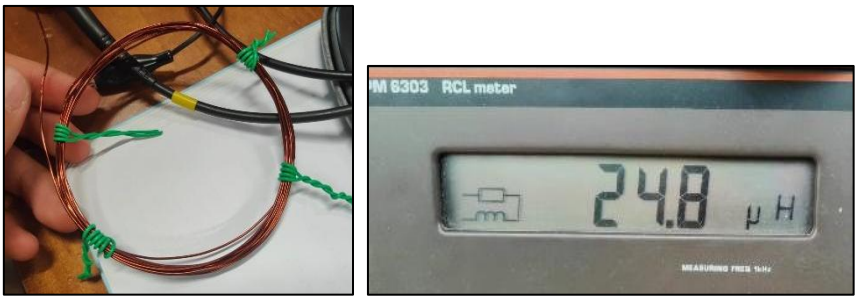


Fig 3.4: Old Antenna Coil and Inductance

$$Required\ C = \frac{3.1979 \times 10^{-12}}{24.8 \times 10^{-6}} = 128nF$$

The initial coil created by the sensors team was suitable for early testing but had relatively low inductance. This limited the detection range of the antenna coil, and we could only obtain usable inputs when the duck was directly above it. However, this would require lifting the duck for reliable sensing, introducing unnecessary mechanical complexity.

Hence, the sensors team created a new coil antenna to solve this issue.

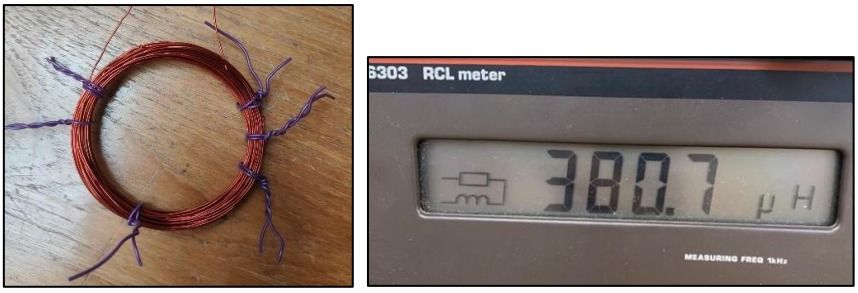


Fig 3.5: New Antenna Coil and Inductance

$$Required\ C\ =\ \frac{3.1979\times10^{-12}}{380.7\times10^{-6}} = 8.4nF$$

The second coil design featured a significantly larger and thicker construction, resulting in approximately 15 times higher inductance compared to the initial version. This improvement enabled reliable signal detection even when the coil was merely in the vicinity of the duck, effectively addressing the limitations of the first design. The system was also able to produce readings consistent with those obtained in direct alignment, without requiring additional mechanical adjustments. We then adjusted the capacitor values as shown above.

Based on the circuit design shown above in Fig 3.2, the circuit in Fig 3.6 was built, mainly consisting of the LCR circuit, the gain stage, the rectification, and the demodulation stage.

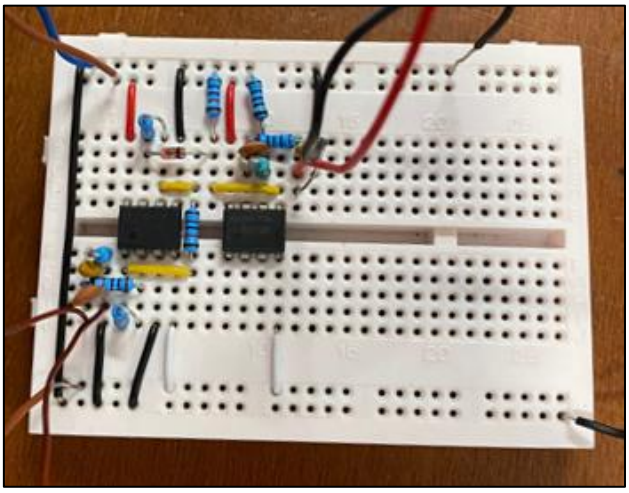


Fig 3.6: Radio Sensor Circuit

The following results in Fig 3.7 were then obtained.

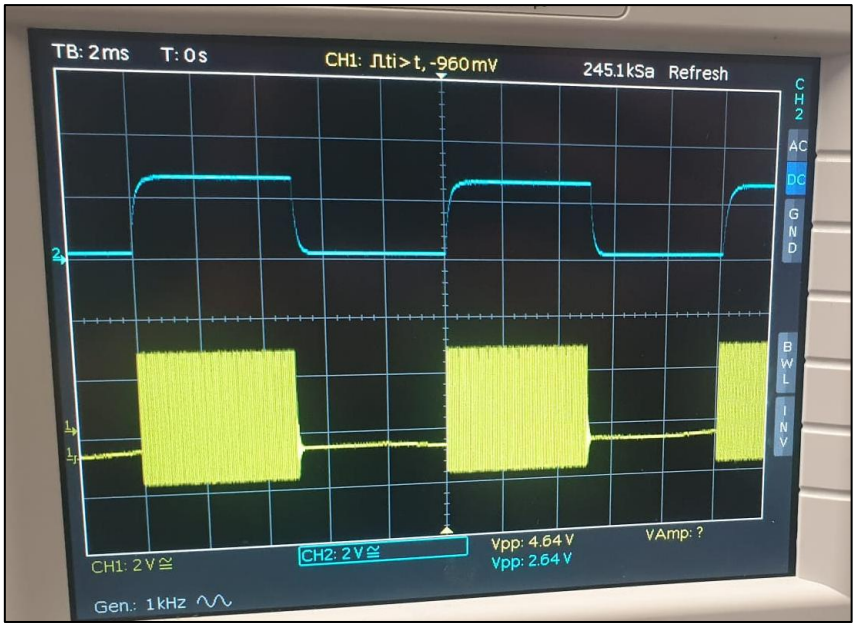


Fig 3.7: Radio Sensor Circuit Output

With these results, the team noticed that the output's rising edge was steep enough that a comparator was not required for the microcontroller to process the results accurately.

3.3 Testing and Characterisation

Test 1: Determination of Maximum Range for Accurate Detection

The first test aimed to characterise how the strength of the radio signal detected by the coiled antenna changes with distance, to determine the maximum range for accurate detection. The duck emitting the radio waves was placed at different distances from the coiled antenna, and the amplitude of the radio signal was recorded accordingly.

Distance (cm)	Amplitude (V)
0	3.68
1	2.4
2	1.68
3	1.2
4	0.96
5	0.72
6	0.6
7	0.52

8	0.36
9	0.32
10	0.312

Fig 3.8: Radio Sensor Test 1 Results

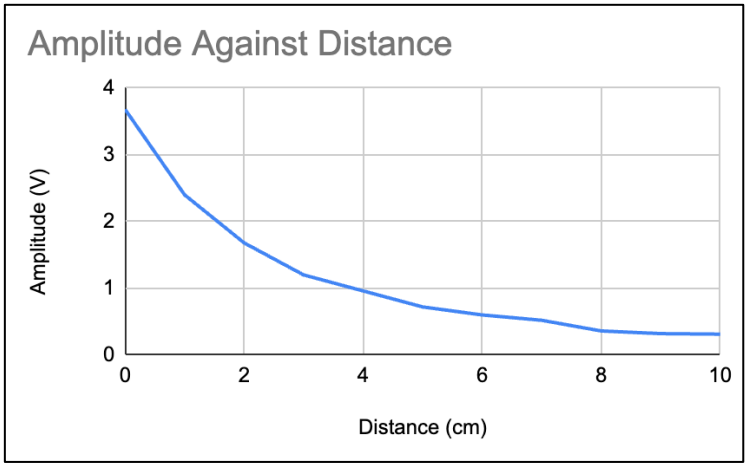


Fig 3.9: Radio Sensor Graph of Amplitude Against Distance

It is observed that the amplitude of the radio signal decreases as the distance of the duck from the coiled antenna increases. This is due to the attenuation of the radio signal over distance. Thus, with this test, the maximum range for accurate detection can be established.

Test 2: Characterisation of Antenna Coil Features

The second test aimed to characterise the main features of our inductor coil. The first of which is the inductance, which is important to determine the capacitor for the resonant circuit. The second of which is the induced voltage, which is important to determine how much amplification is required.

The main equations involved in the inductor coil are $L = \frac{\mu N^2}{l} = \frac{\mu N}{2\pi r}$ and $E = -N\frac{d\Phi}{dt}$ (Faraday's Law), where L is the inductance in H, μ is the permittivity of free space, N is the number of turns of the coil, l is the length of the coil and equal to $2\pi r$N, E is the induced electromotive force (EMF), and $\frac{d\Phi}{dt}$ is the rate of change of magnetic flux.

The number of turns was first fixed to 5 turns, while the radius of the coil was varied. We used the RLC Bridge to measure the respective coil inductances, and the capacitance value for resonance was then calculated accordingly, and the amplitude of the radio signal detected by the coiled antenna was recorded.

Radius of Coil (cm)	Inductance (µH)	Resonant Capacitance (nF)	Amplitude (mV)
6	4.1	780	11
5	5	630	8
4	5.9	542	5
3	6.5	497	3
2	6.9	463	0.6

Fig 3.10: Radio Sensor Test 2.1 Results

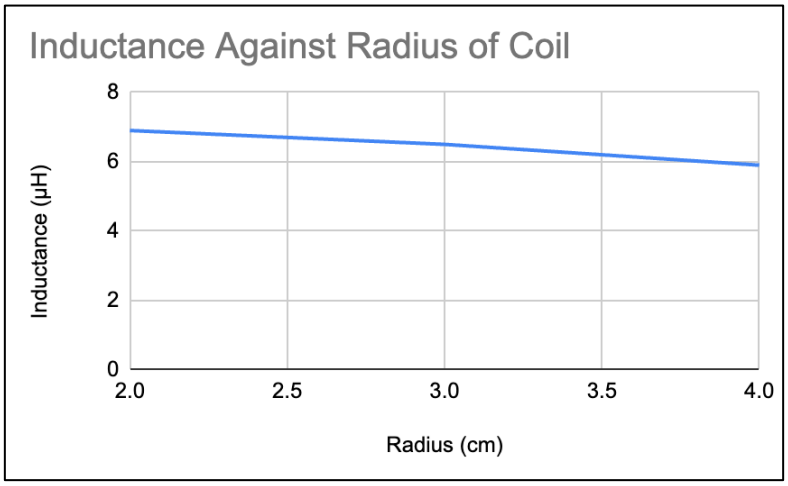


Fig 3.11: Radio Sensor Graph of Inductance against Radius of Coil

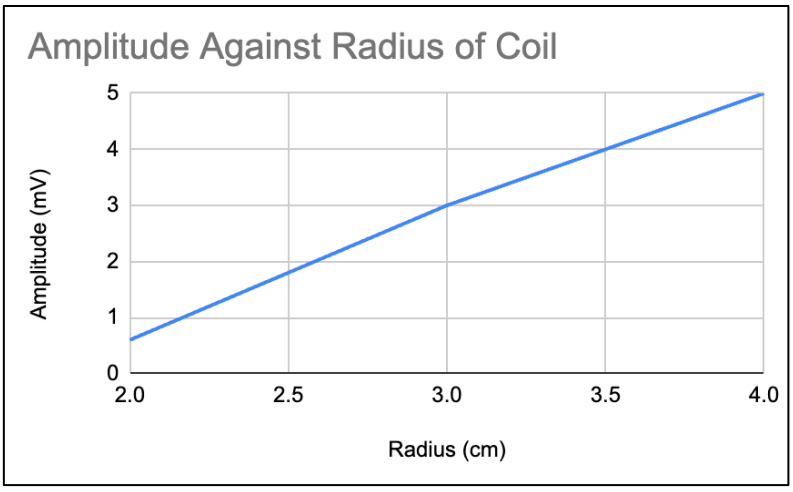


Fig 3.12: Radio Sensor Graph of Amplitude Against Radius of Coil

It is observed that the inductance of the coiled antenna decreases as its radius increases. This can be seen in the equation $L = \frac{\mu N^2}{l}$, since l it increases when the radius increases, and the inductance is inversely proportional to the length of the coil. It is also observed that the amplitude of the induced voltage in the coiled antenna increases as the radius increases. This is due to how the area of the coil increases, which increases the induced voltage in the coil by Faraday's Law.

The radius was then fixed to 4cm while varying the number of turns in the coiled antenna. Similarly, we used the RLC Bridge to measure the respective coil inductances, and the capacitance value for resonance was then calculated accordingly. The amplitude of the radio signal detected by the coiled antenna was then recorded.

Number of Turns of Coil	Inductance (µH)	Resonant Capacitance (nF)	Amplitude (mV)
9	10.8	296	20
8	9.7	330	13
7	8.3	385	9
6	7.2	444	7
5	5.9	542	5
4	4.4	727	4.1
3	3.8	842	3.5
2	3.1	1.03	2.2
1	2.9	1.1	1.25

Fig 3.13: Radio Sensor Test 2.2 Results

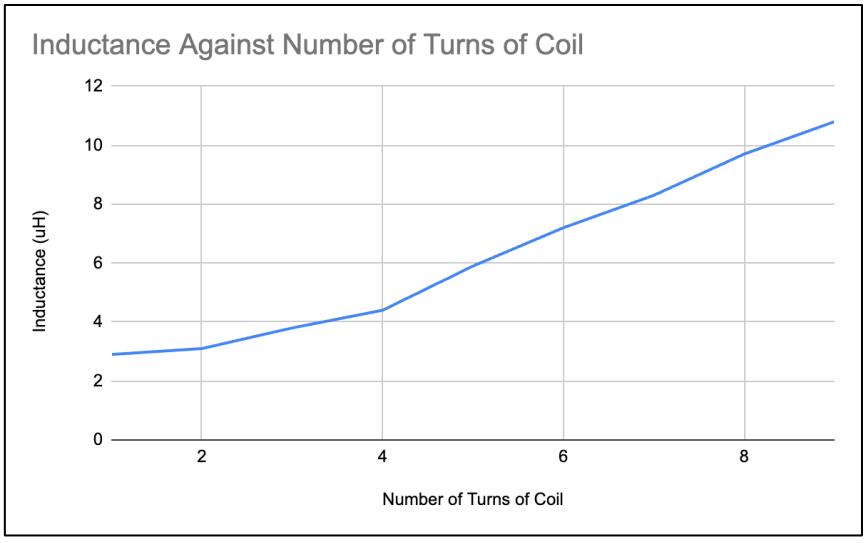


Fig 3.14: Radio Sensor Graph of Inductance against Number of Turns of Coil

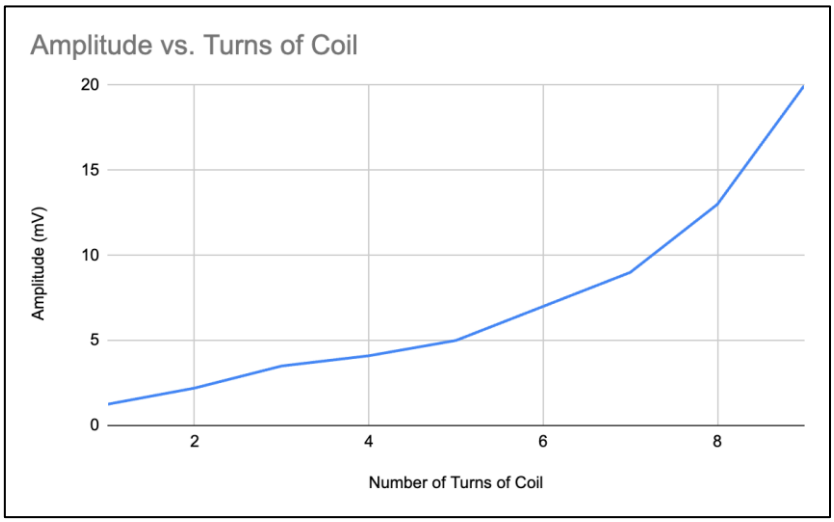


Fig 3.15: Radio Sensor Graph of Amplitude against Number of Turns of Coil

It is then observed that the inductance of the coil increased as the number of turns of the coiled antenna increased. This can be seen in the equation $L = \frac{\mu N}{2\pi r}$, where the inductance is directly proportional to N. Furthermore, it is also observed that the amplitude of the induced voltage in the coiled antenna increases as the number of turns of the coiled antenna increases. This can be seen in the equation $E = -\frac{N d\Phi}{dt}$, where the induced EMF is directly proportional to N.

Hence, to make a coiled antenna that would give high induced voltage and high inductance, the coiled antenna would need a moderately large radius with many turns. In the circuit built by the sensors team, the coil has a radius of 4cm and 40 turns, which satisfy the above requirements.

Test 3: Integrated Testing with App UI

The final test aimed to ensure the whole subsystem works. The team tested the radio coil antenna and circuit integrated with the code and app, and managed to consistently obtain the results required, as shown in Fig 3.16.

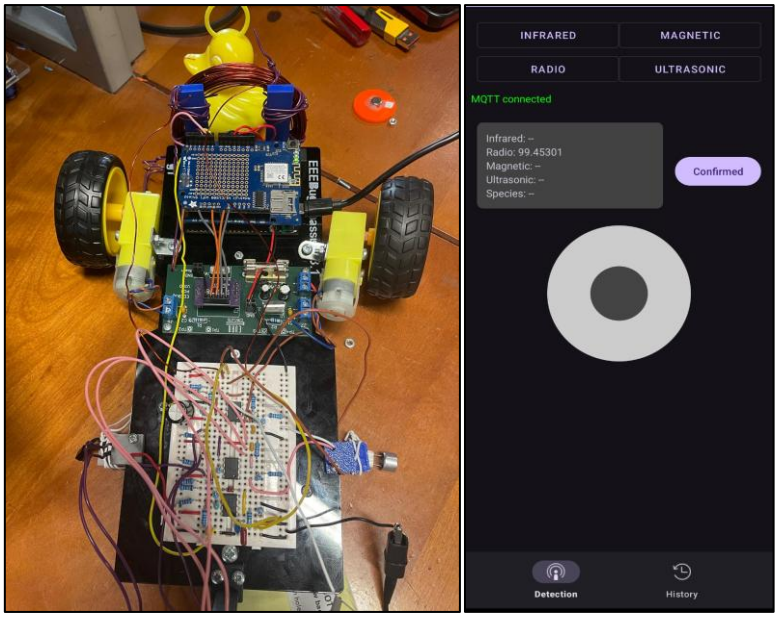


Fig 3.16: Radio Sensor Integrated Testing

4 Magnetic Field Sensor

4.1 Design Overview

In this subsystem, the task is to detect either a downward or an upward magnetic field, each of which produces a different output DC voltage value.

This is implemented by placing a Hall effect sensor parallel to the face of the magnet. The voltage obtained from the Hall effect sensor is amplified with a non-inverting amplifier. The code then compares the amplified voltage to a suitable threshold to determine whether the magnetic field is upwards or downwards. The detected voltage is negative when the magnetic field is upwards and positive when the magnetic field is downwards.

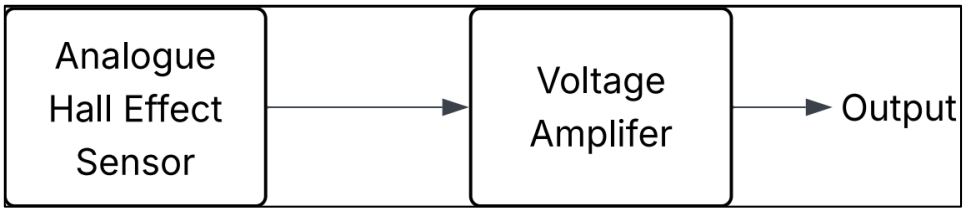


Fig 4.1: Magnetic Field Sensor System Flowchart

The magnetic field of the duck is static, which means that it is unable to induce a current in a stationary coil of wire. Hence, the sensors team decided to source and purchase a linear Hall effect sensor to detect the magnetic field direction of the duck. The team found a low-noise linear Hall effect sensor with an analogue output, the A1324LUA-T. It is a rail-to-rail radiometric sensor that gives a voltage output proportional to the magnetic field strength with a sensitivity of 5.0mV/G. It has a quiescent voltage output of 50% of the supply voltage, which in our setup, which has a range of -3.3V to 3.3V, would be around 0V. It also has a suitably wide ambient operating temperature range of -40 ºC to 150 °C, ensuring stable operation.

The presence of a downward magnetic field perpendicular to the branded face of the sensor would increase the output voltage to become more positive, while an upwards magnetic field decreases the output voltage to become more negative. The magnetic field might be weak, and as such, the output voltage may be small. Hence, there is a need to amplify the voltage output accordingly. With these results and the sensor, the team has chosen, the team has the following circuit design in Fig 4.2.

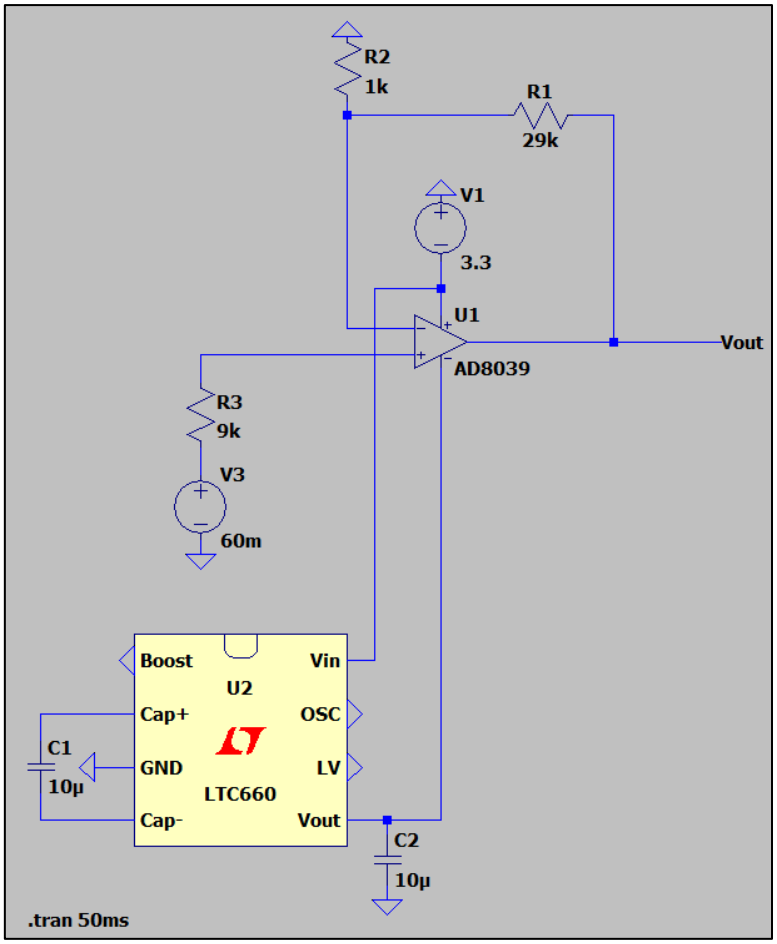


Fig 4.2: LTspice Diagram for Magnetic Field Sensor Circuit

The circuit includes a non-inverting amplifier with a gain of 30, and the output is then piped to the microcontroller as an analogue input for processing.

4.2 Technical Solutions and Build

The typical supply voltage of the A1324LUA-T sensor is 3.3V, which makes it very suitable for the circuit, which mainly runs on 3.3V power supply. The DC output voltage range is also large, from 0.33V to 2.97V, which provides a large enough range for the microprocessor's code to process accurately.

Based on the circuit design shown above in Fig 4.2, the draft circuit below was built, mainly consisting of the sensor and the gain stage. Based on initial tests, with the sensor 2 cm from the magnet inside the duck, the final output voltage after amplification will give 2.6V with a downwards magnetic field, and -1.8V with an upwards magnetic field.

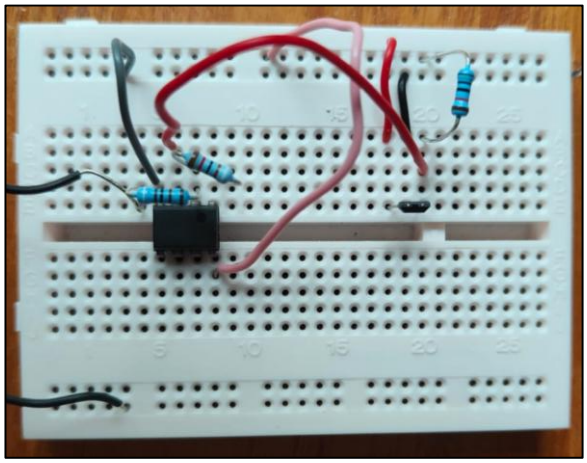


Fig 4.3: Magnetic Field Sensor Circuit

The results below were obtained with the magnetic field sensor circuit.

For the quiescent voltage output, there is a slight offset for the output that is amplified as shown in Fig 4.4. However, the mechanical and UI team took that into account for the processing of the output, and this offset will not affect the results that are sent to the user on the app.

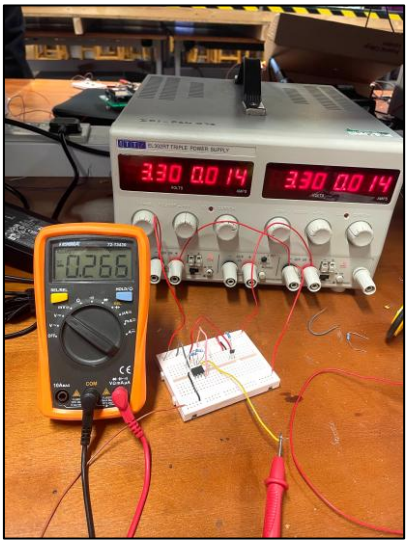


Fig 4.4: Output Voltage for Neutral Magnetic Field

Fig 4.5 shows the amplified voltage output for an upwards magnetic field, with a voltage output of roughly -1.8V.

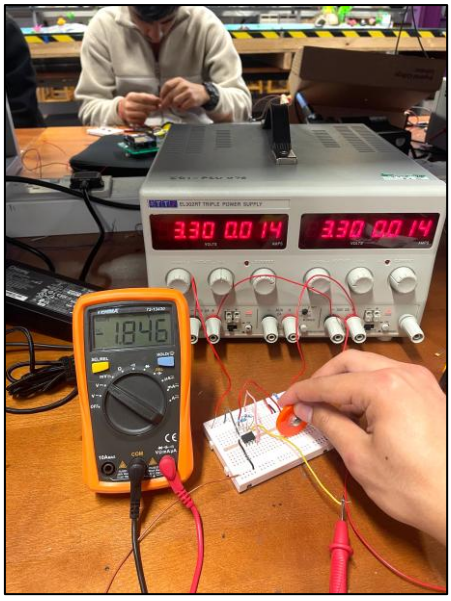


Fig 4.5: Output Voltage for Upwards Magnetic Field

Lastly, the Fig 4.6 shows the amplified voltage output for a downwards magnetic field, with a voltage output of roughly 2.6V.

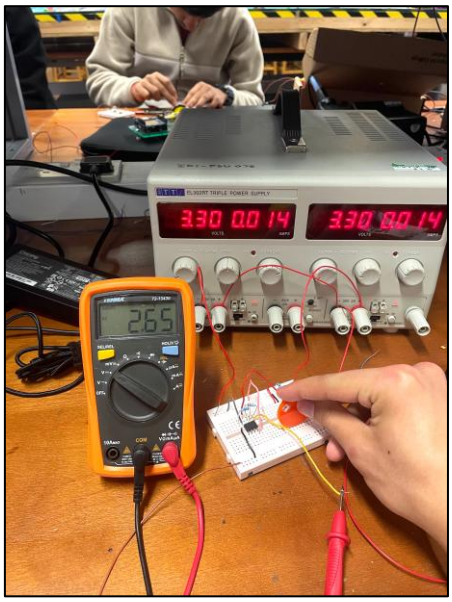


Fig 4.6: Output Voltage for Downwards Magnetic Field

In the tests conducted, the sensors team noticed that to obtain accurate results, the sensor's face must be parallel to that of the magnet. This was a very important discovery, as the sensor gives an inaccurate output if the magnetic field is not perpendicular to that of the sensor's face. This

information was passed on to the mechanical and UI teams for them to take into consideration for the design of the rover.

4.3 Testing and Characterisation

Test 1: Determination of Maximum Range and Orientation for Accurate Detection

The first test conducted aimed to verify the upwards or downwards polarity detection of the sensor and circuit. This is achieved by placing magnets with known polarities near the sensor and measuring the respective output voltages. The results obtained are shown in Fig 4.7.

Distance (cm)	Output Voltage (V)	Magnetic Field Direction
0	-3	Up
1	-2.9	Up
2	-2.07	Up
3	-1.11	Up
4	-0.93	Up
5	-0.84	Up

Fig 4.7: Magnetic Field Sensor Test 1 Results (1)

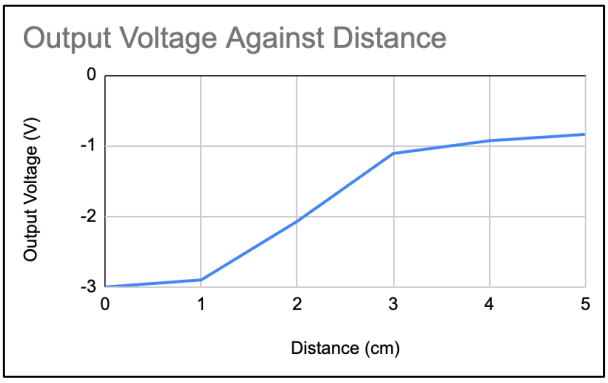


Fig 4.8: Magnetic Field Sensor Graph of Output Voltage Against Distance (1)

Distance (cm)	Output Voltage (V)	Magnetic Field Direction
0	3.1	Down
1	2.7	Down
2	1.62	Down
3	0	Down
4	-0.3	Up
5	-0.51	Up

Fig 4.9: Magnetic Field Sensor Test Results (2)

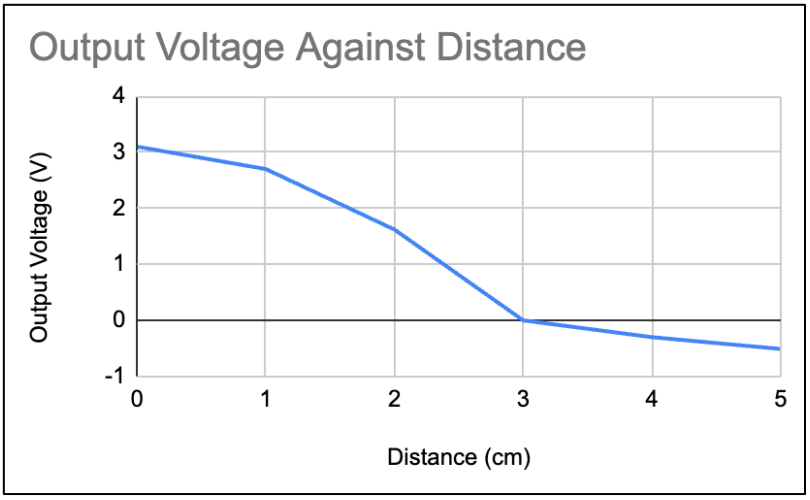


Fig 4.10: Magnetic Field Sensor Graph of Output Voltage Against Distance (2)

A negative voltage indicates an upward magnetic field, while a positive voltage indicates a downward magnetic field. Initially, when the magnetic field sensor is touching the magnet, the output voltage is saturated at around 3V. As the distance between the magnet and the sensor increases, the magnetic field strength, and thus the voltage magnitude, decreases. This is so since magnetic fields weaken with distance from the source.

However, the team noticed that if the magnet is too far away from the sensor, the results could become inaccurate, as seen from the second table of results, since the magnetic field strength would be too weak. Hence, the team determined that an optimal sensing distance would be 2cm from the magnet, and this information was passed over to the mechanical and UI teams for consideration.

Test 2: Integrated Testing with App UI

The final test aimed to ensure the whole subsystem works. The team tested the magnetic field sensor and circuit integrated with the code and app, and managed to consistently obtain the results required, as shown in Fig 4.9.

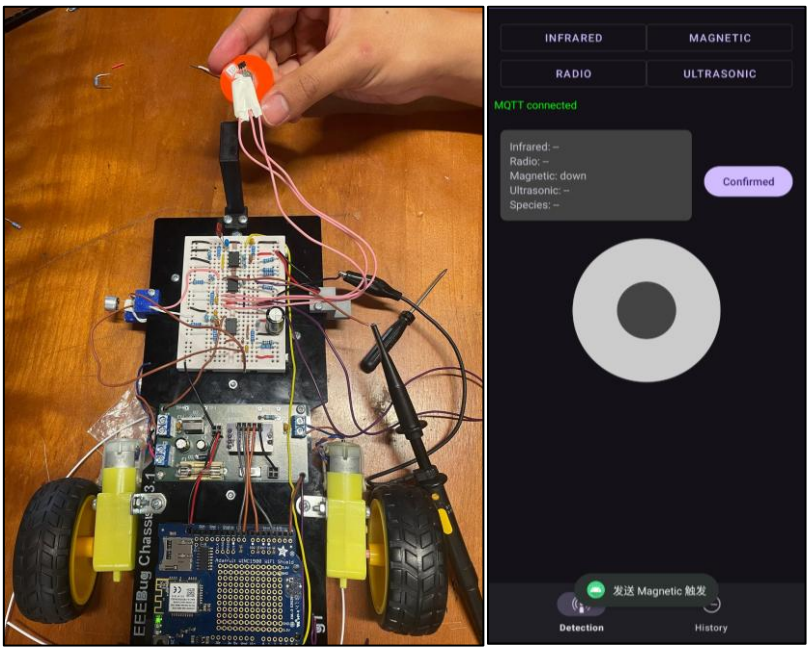


Fig 4.9: Magnetic Field Sensor Integrated Testing

5 Infrared Sensor

5.1 Design Overview

In this subsystem, the task is to measure the pulse frequency of an infrared signal emitted by the duck.

To execute this, an infrared sensor is used to measure the signal, and its output is amplified with a non-inverting amplifier, ensuring that the signal is large enough to be processed accurately. The code then determines the pulse frequency emitted to determine the species of the duck.

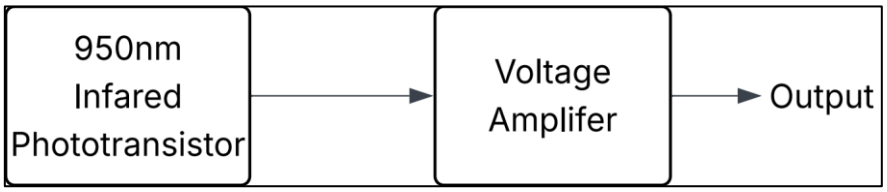


Fig 5.1: Infrared Sensor System Flowchart

Ambient light will cause interference to the sensor, and hence the sensors team found the BPW82 photodiode with high speed and high radiant sensitivity. The sensor has an in-build daylight blocking filter and a spectral bandwidth range of 790nm to 1050nm, having peak sensitivity at 950nm, which is the wavelength of infrared signals. With this in-built filter, we can ensure that other sources of light from the environment do not interfere with measurements. The photodiode also has a high angle of half sensitivity at 65°, which allows it to be less directional and have a wider field of view. This reduces restrictions for the mechanical and UI team when designing where to place the sensor. It also has a suitably wide ambient operating temperature range of -40 ºC to 100 ºC, ensuring stable operation.

The frequency range of the pulses is 135–1000Hz, but the pulse width of the infrared signal is 50µs and the amplitude observed will be reduced if the high frequency harmonics are filtered. Low-pass filtering may smooth out the edges of the pulse, reducing the accuracy of the measurement. Since the photodiode has a low diode capacitance with a maximum of 70pF, thus low pass filtering effects are not significant for detecting the pulse frequencies which are given for the duck species.

With the sensor the team has chosen, the team has the following circuit design in Fig 5.2 and 5.3.

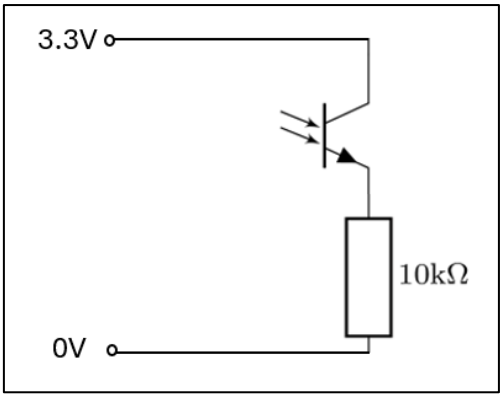


Fig 5.2: Infrared Sensor Circuit Diagram

The infrared sensor is set up as shown above in Fig 5.2, and its output is then connected to the circuit below in Fig 5.3.

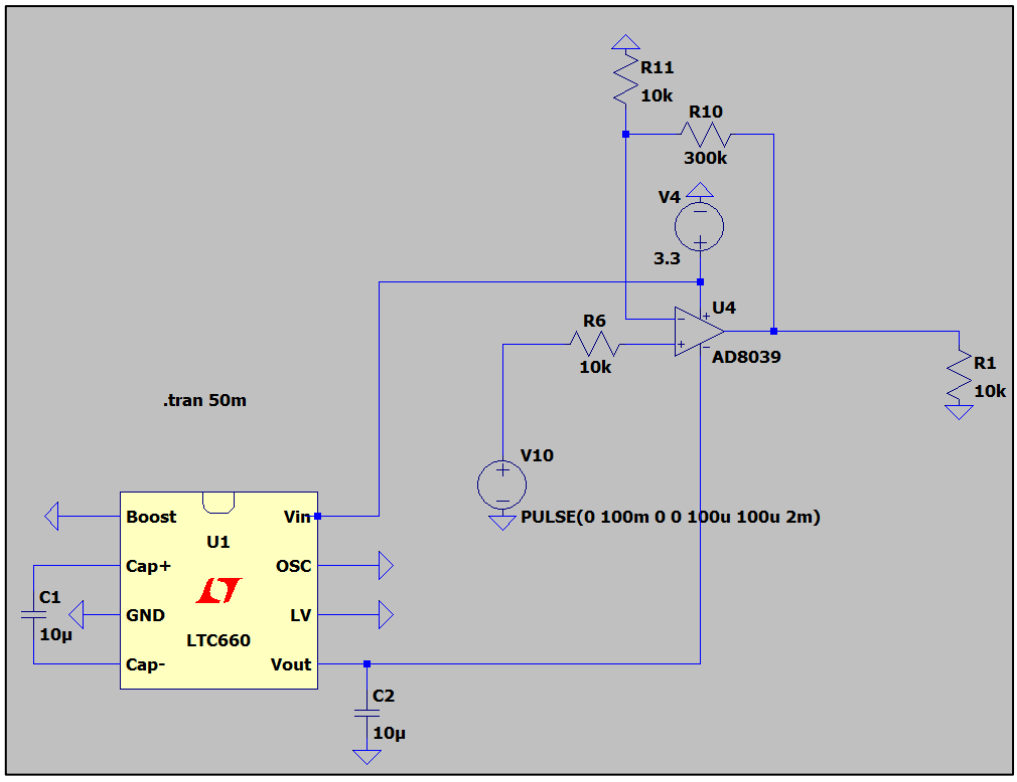


Fig 5.3: LTspice Diagram for Infrared Sensor Circuit

The detected infrared signal pulses are amplified using a non-inverting amplifier with a gain of 30, and their output is then piped to the microcontroller as an analogue input for processing.

After simulating the circuit on LTspice, the following output in Fig 5.4 is obtained.

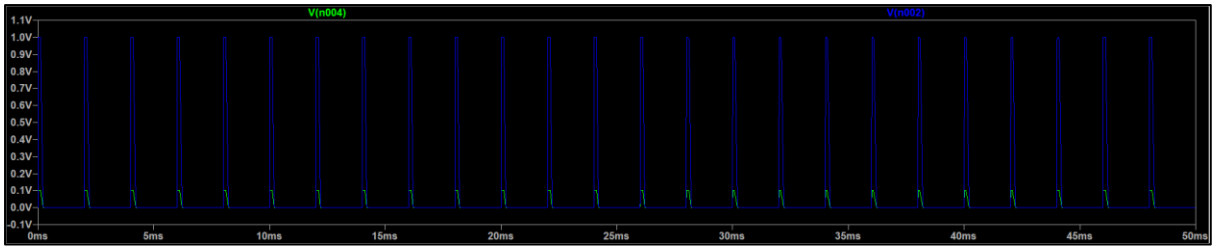


Fig 5.4: LTspice Simulation Output for Infrared Sensor Circuit

5.2 Technical Solutions and Build

Based on the circuit design shown above in Fig 5.2 and 5.3, the draft circuit below in Fig 5.5 was built, mainly consisting of the sensor and the gain stage. In this build, the SFH300 phototransistor from the Spring Term Lab Skills section was used for testing.

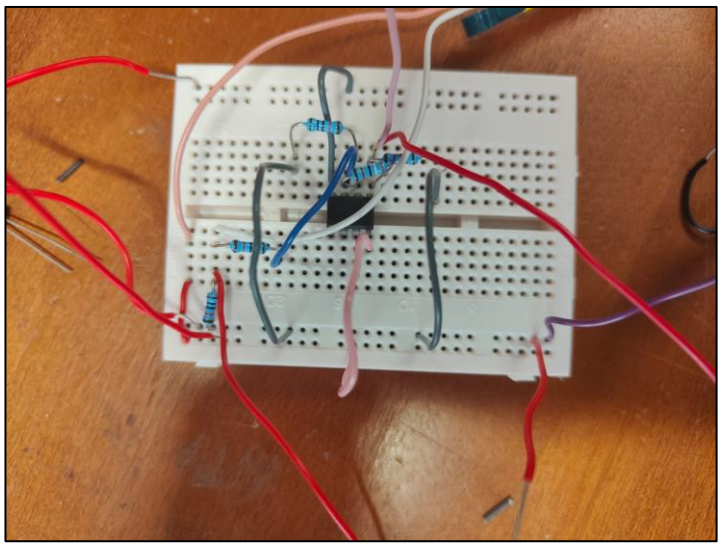


Fig 5.5: Initial Infrared Sensor Circuit

Based on initial tests, with the sensor 2 cm from the duck, the final output voltage of the pulses after amplification will be around 2V.

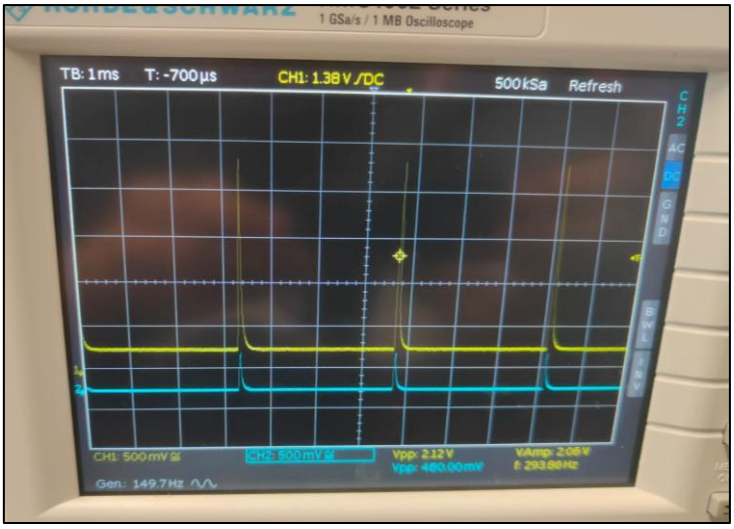


Fig 5.6: Initial Infrared Sensor Circuit Output

After significant testing, it can be concluded that ambient light does indeed cause interference with the SFH300 phototransistor by causing an offset to the voltage output. This will affect the accuracy of the results given by the microcontroller after processing. This issue was tackled in a two-pronged approach. The first of which would be to use the new BPW82 photodiode, which would filter out visible light from the environment, as mentioned previously. The second would be to change the code to detect based on the change in voltage rather than simply set a threshold voltage for comparison. With the above testing and results, the new circuit below in Fig 5.7 was built with the new BPW82 photodiode.

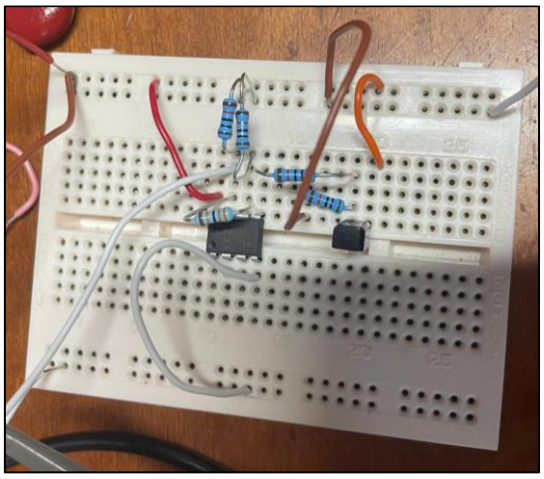


Fig 5.7: Final Infrared Sensor Circuit

Testing was done with the new circuit, obtaining the results as shown in Fig 5.8.

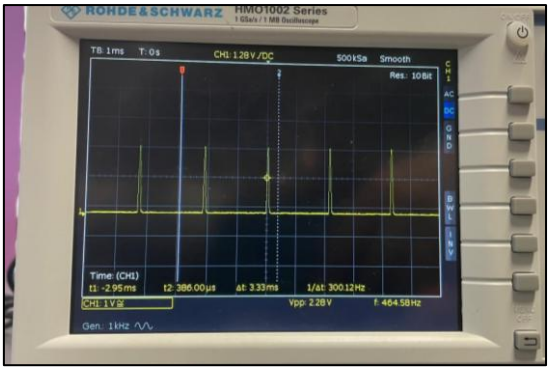


Fig 5.8 Final Infrared Sensor Circuit Output

5.3 Testing and Characterisation

Test 1: Determination of Maximum Range for Accurate Detection

The first test aimed to characterise how the strength of the infrared pulse signal detected by the BPW82 photodiode changes with distance, to determine the maximum range for accurate detection. In this test, the IR LED was modulated at the target frequency, and the amplitude of the demodulated signal was recorded at each distance. The following results in Fig 5.9 were obtained.

Distance (cm)	Amplitude (V)
0	2.88
1	2.64
2	1.6
3	1.52
4	1.2
5	0.88
6	0.64
7	0.56
8	0.48
9	0.4
10	0.4
11	0.4
12	0.32
13	0.32
14	0.32
15	0.32

Fig 5.9: Infrared Sensor Test 1 Results

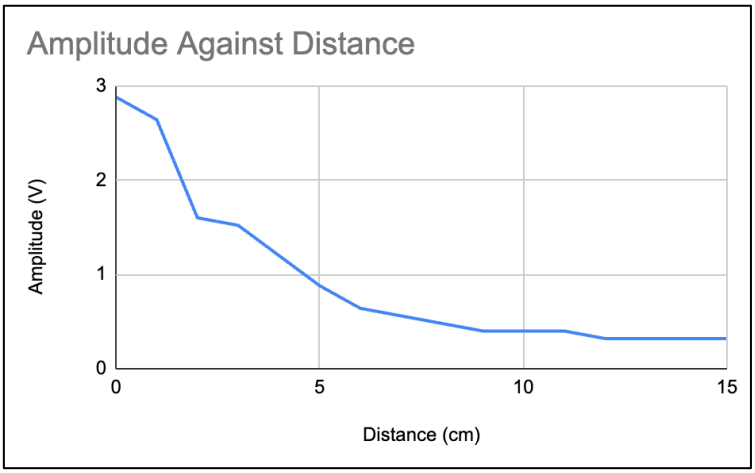


Fig 5.10: Infrared Sensor Graph of Amplitude Against Distance

The recorded amplitude decreases with increasing distance between the sensor and the duck. This is due to the attenuation of the infrared pulse signals over distance, which reduces the amount of reflected light reaching the sensor. Thus, with this test, the maximum range for accurate detection can be established.

Test 2: Evaluation of In-Built Filtering Capabilities

The second test aimed to evaluate the new BPW82 photodiode filtering capabilities. The sensors team tested both the old SFH300 phototransistor and the new BPW82 photodiode with three brightness levels. The results obtained are as shown in Fig 5.11.

Brightness	**Light Sensor/mV**	**New Infrared Sensor/mV**
Darkness	-20	280
Room Lighting	87.5	280
Direct Light	200	280

Fig 5.11: Infrared Sensor Test 2 Results

The SFH300 phototransistor showed changes in its output voltage in response to varying ambient light levels. In contrast, the BPW82 photodiode had a consistent output voltage regardless of ambient light, indicating that the in-built filter was working well to filter out light of wavelengths other than that of infrared.

Test 3: Integrated Testing with App UI

The final test aimed to ensure the whole subsystem works. The team tested the infrared sensor and circuit integrated with the code and app, and managed to consistently obtain the results required, as shown in Fig 5.12.

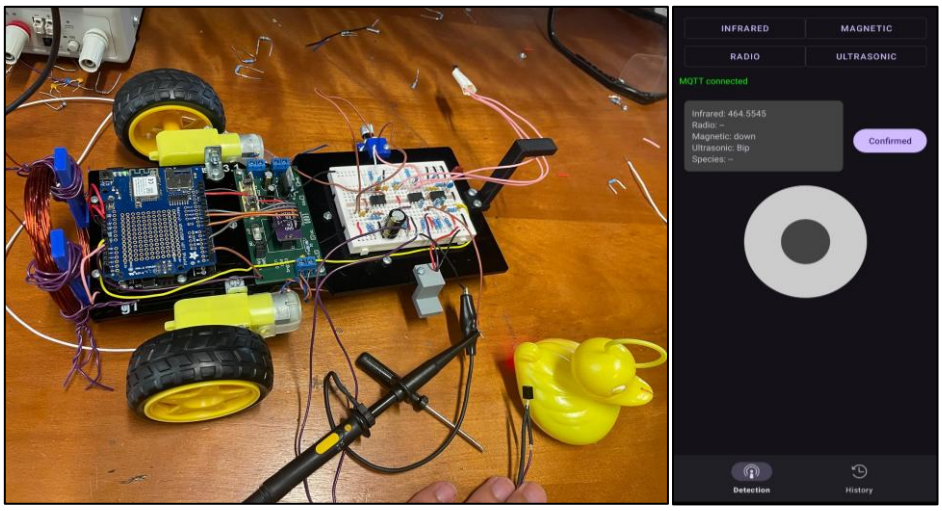


Fig 5.12: Infrared Sensor Integrated Testing

6 Ultrasonic Sensor

6.1 Design Overview

In this subsystem, the task is to detect the ultrasonic waves emitted by the duck and decode it to obtain its name.

To execute this, an ultrasonic transducer is used to receive the ultrasonic waves, and its output will then have to be amplified before the signal is demodulated. The demodulation involves rectification of the amplified signal before passing it through a low-pass filter.



Fig 6.1: Ultrasonic Sensor System Flowchart

Fig 6.2 shows how the duck's name is encoded in the ultrasonic waves.

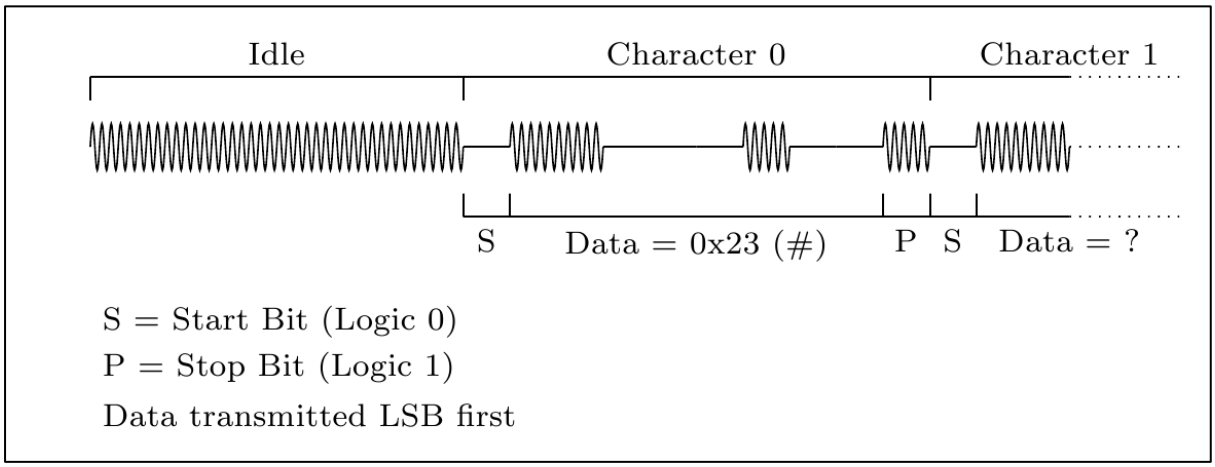


Fig 6.2: UART Encoding

The communication between the duck and the ultrasonic transducer uses a carrier frequency of 40kHz, modulated with two-level amplitude-shift keying. The name is encoded using American Standard Code for Information Interchange (ASCII) character codes in Universal Asynchronous Receiver/Transmitter (UART) packets with 1 start bit (logic 0) and 1 stop bit (logic 1). The data rate is 600 bits per second, and each duck's name is four characters long, including an initial "#" symbol (0x23).

The sensors team found an ultrasonic receiver, the 400SR100, which has a centre frequency of 40kHz. This ensures maximum sensitivity and response to the intended ultrasonic signals while rejecting out-of-band noise or other frequencies. It also has a bandwidth of 3kHz around 40kHz, which is narrow enough to filter out noise, but wide enough to allow for slight frequency variations. The receiver has a wide acceptance angle of about 72°, which results in

it being able to detect ultrasonic signals from a broad range of directions, reducing the restrictions for the mechanical and UI team. Lastly, it also has a suitably wide ambient operating temperature range of -30 ºC to 70 ºC, ensuring stable operation. With this data and the sensor, the team has chosen, the team has the following circuit design in Fig 6.3.

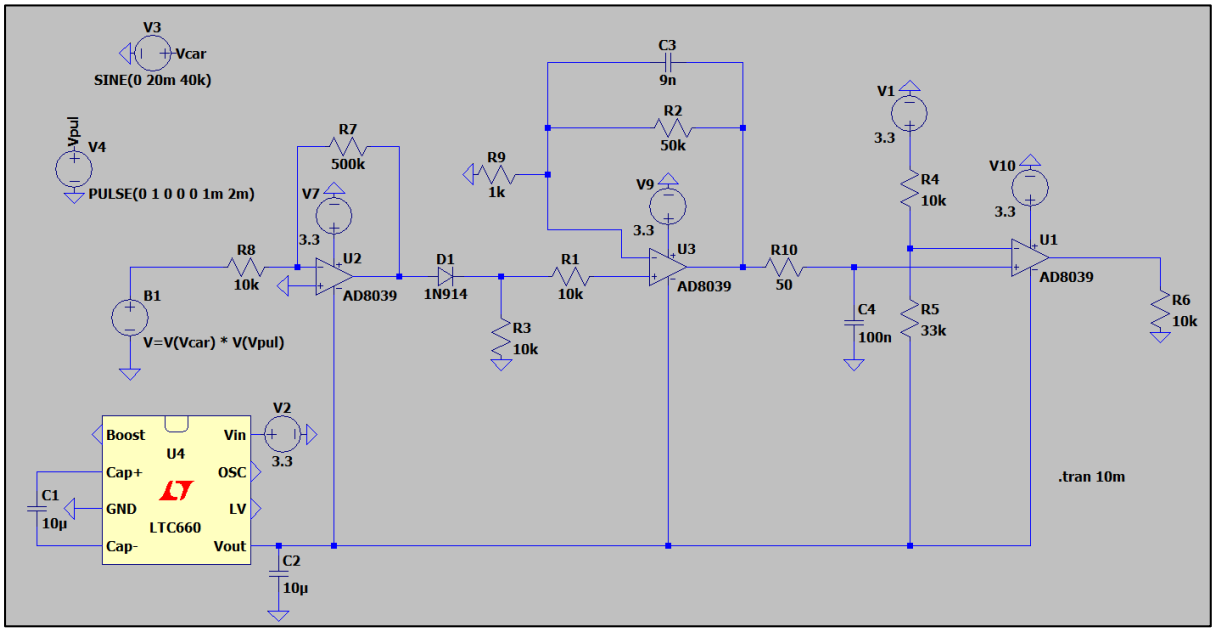


Fig 6.3: LTspice Diagram for Ultrasonic Sensor Circuit

The ultrasonic receiver picks up the signal emitted by the duck, and the output is then amplified using an inverting amplifier gain of 50, which is then rectified and demodulated. Afterwards, the demodulated output is put into a comparator which compares the amplitude to a threshold voltage of 2.5V to convert the analogue voltage values into a binary signal. The final output after the comparator is then piped to the microcontroller as a digital input for processing.

After simulating the circuit on LTspice, the following output in Fig 6.4 is obtained.

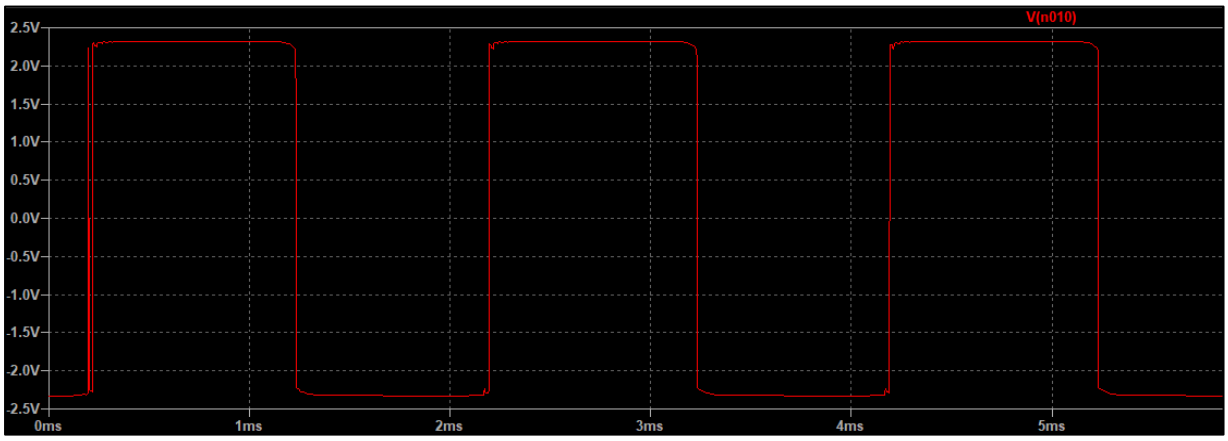


Fig 6.4: LTspice Simulation Output for Ultrasonic Sensor Circuit

When the ultrasonic signal is received by the sensor, the sensor then sends the encoded ASCII characters to the serial port receiving end (RX, pin PIN0) of the main control board through the serial port (TX) at a baud rate of 600.

6.2 Technical Solutions and Build

Based on the circuit design shown above in Fig 6.3, the circuit in Fig 6.5 and 6.6 was built. Fig 6.5 shows the sensor, the gain stage, the rectification and demodulation stage, while Fig 6.6 shows the comparator stage. Based on initial tests, with the sensor 2 cm from the duck, the final output voltage will be about 3V.

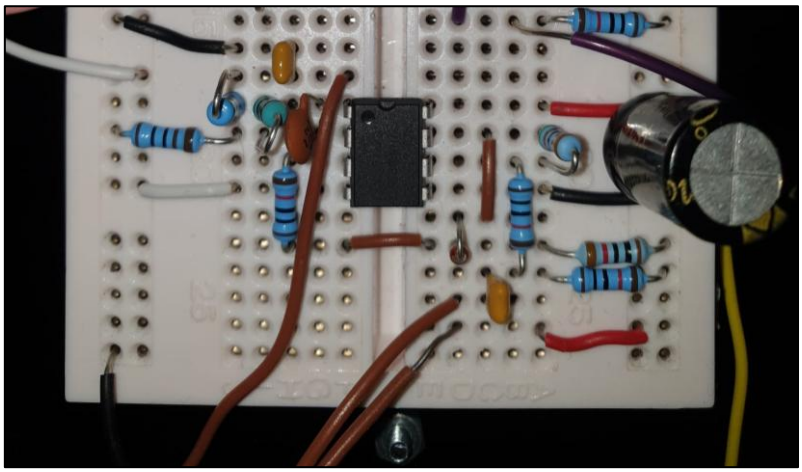


Fig 6.5: Ultrasonic Sensor Circuit (1)

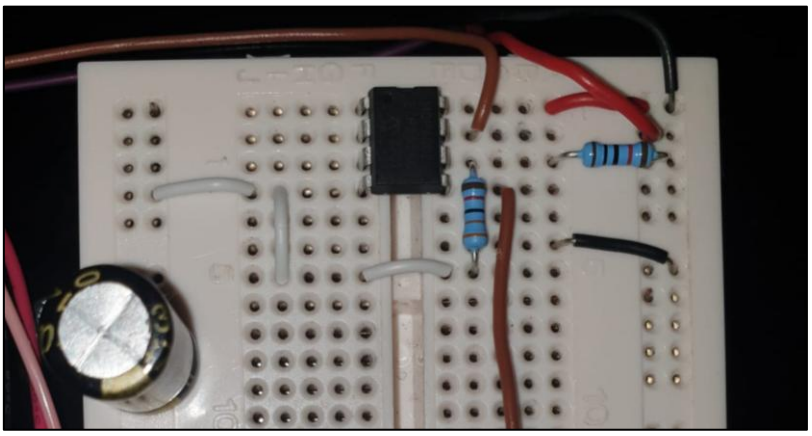


Fig 6.6: Ultrasonic Sensor Circuit (2)

After significant testing, we managed to consistently obtain the following results, which clearly show the 4x8 bytes that form the duck's name, including the initial "#" symbol.

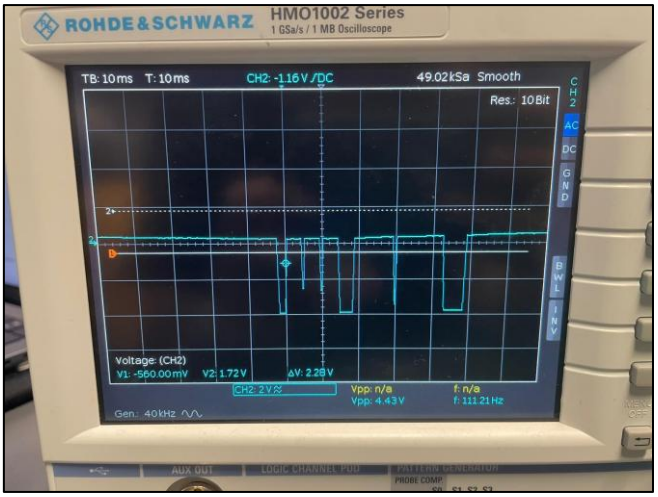


Fig 6.7: Ultrasonic Sensor Circuit Output

6.3 Testing and Characterisation

Test 1: Determination of Maximum Range for Accurate Detection

The first test aimed to characterise how the strength of the ultrasonic wave signal detected by the receiver changes with distance, to determine the maximum range for accurate detection. The duck emitting the ultrasonic waves was placed at different distances from the receiver, and the amplitude of the demodulated signal was recorded accordingly.

Distance (cm)	Amplitude (V)
0	3.5
1	2.68
2	2.44
3	1.52
4	1.44
5	1.32
6	1.2
7	1.1
8	0.9
9	0.7
10	0.5

Fig 6.8: Ultrasonic Sensor Test 1 Results

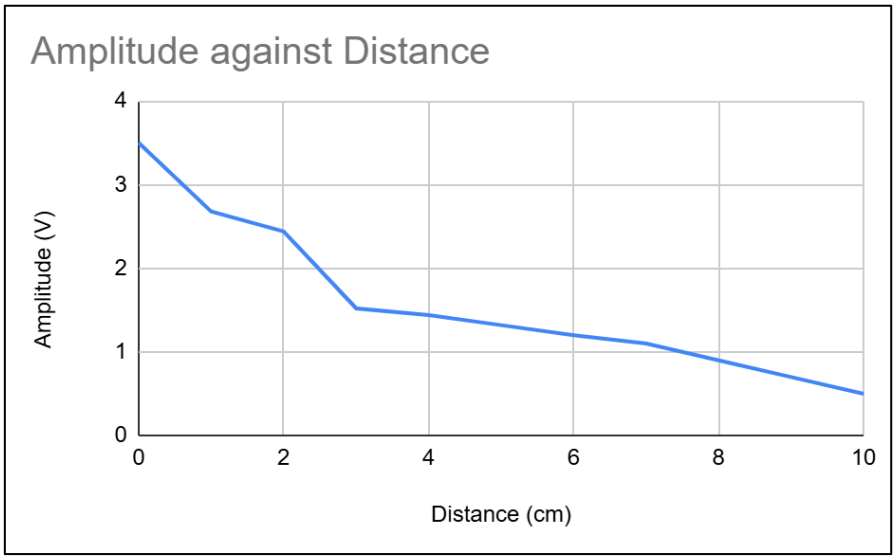


Fig 6.9: Ultrasonic Sensor Graph of Amplitude Against Distance

It is observed that generally, as the distance increases, the maximum peak amplitude for the voltage output decreases, due to the increased attenuation. Hence, the sensor should be placed as close to the duck as possible, and this information was passed to the mechanical and UI team for their consideration.

Test 2: Determination of Sensor Angle Sensitivity

The second test aimed to determine how the orientation of the ultrasonic receiver with respect to the duck affects the ultrasonic signal strength detected by the receiver. This is done to characterise the sensor's angular tolerance and directional sensitivity. The ultrasonic receiver was rotated in 30° increments relative to the duck, for a total of 180° and the amplitude of the demodulated signal at each angle was recorded.

Sensor Angle (Degrees)	**Amplitude (V)**
-90	0.8
-60	1.5
-30	2.6
0	3
30	2.5
60	1.3
90	1

Fig 6.10: Ultrasonic Sensor Test 2 Results

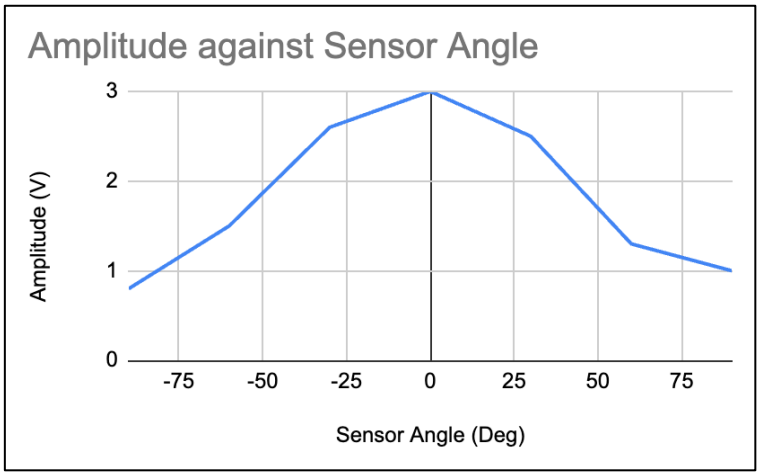


Fig 6.11: Ultrasonic Sensor Graph of Amplitude Against Sensor Angle

It was observed that the output voltage amplitude reached a peak value when the sensor was directly facing the duck and reduced as it was rotated away from the duck. The output voltage was relatively constant between -30º and 30º, since the sensor is supposed to have an acceptance angle of about 72º. As the angle continues to be moved away from this position, the signal strength drops, showing that the ultrasonic sensor has a directional radiation pattern. This information was then passed to the mechanical and UI team for consideration.

Test 3: Integrated Testing with App UI

The final test aimed to ensure the whole subsystem works. The team tested the ultrasonic receiver and circuit integrated with the code and app, and managed to consistently obtain the results required, as shown in Fig 6.12.

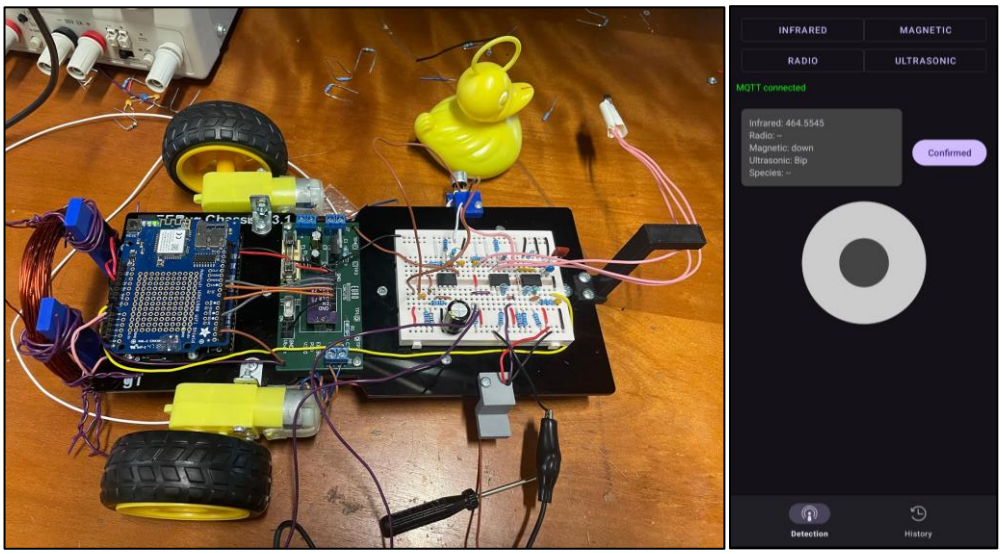


Fig 6.12: Ultrasonic Sensor Integrated Testing

7 Rover Movement and Mechanical Design

7.1 Design Overview

Fig 7.1 shows a flowchart that gives an overview of the code developed to control the rover via the app. This section will focus more on the mechanical aspects of the rover, and the next section will discuss the UI aspect of it.

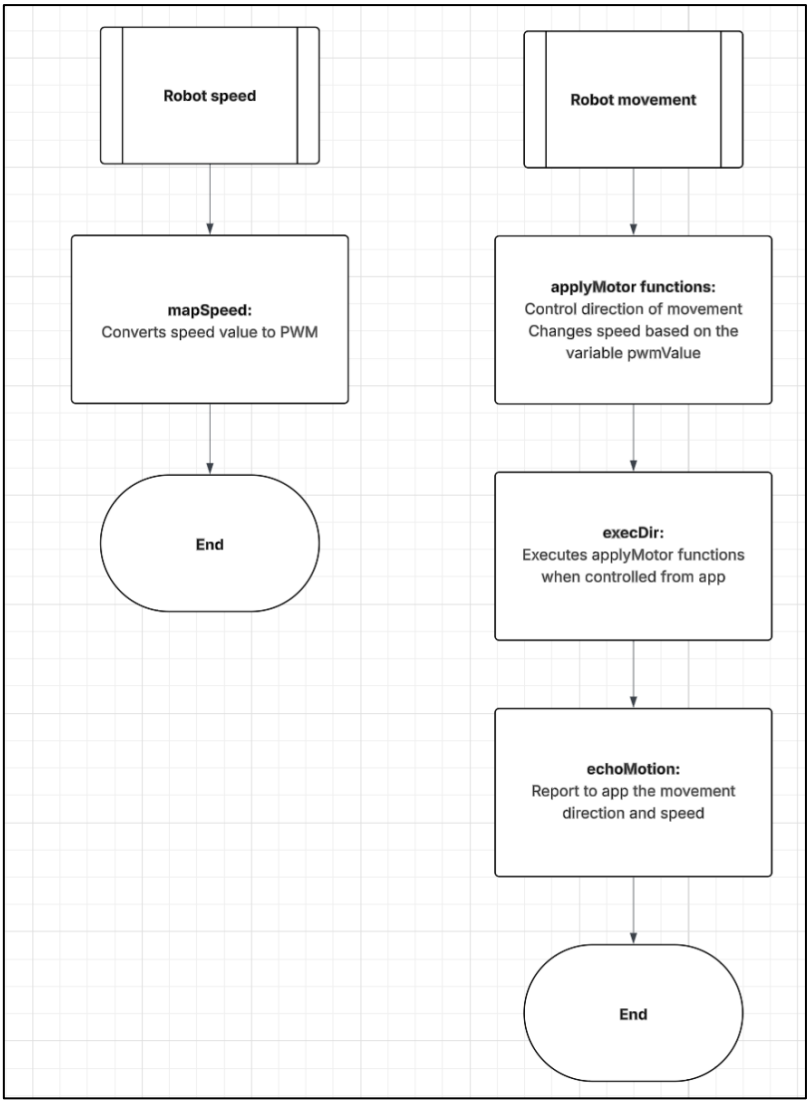


Fig 7.1: Flowchart subroutines for movement and speed

The mechanical and UI team came up with two main designs for the chassis of the rover, considering the considerations provided by the sensors team for the placement and integration of the sensors. Another main consideration would be to keep the rover design as simple as possible. Since the EEEBug chassis was designed to be a useful platform, the concept designs were based on it with some modifications. The weight of the rover was considered

significantly, since there is a weight limit of 750g. Lastly, the size restrictions helped determine the final design for the chassis, since the width of the lakeside entrance was about 250mm, with a maximum height of about 150mm and a minimum height of about 110mm.

The first design concept is as shown in Fig 7.2.

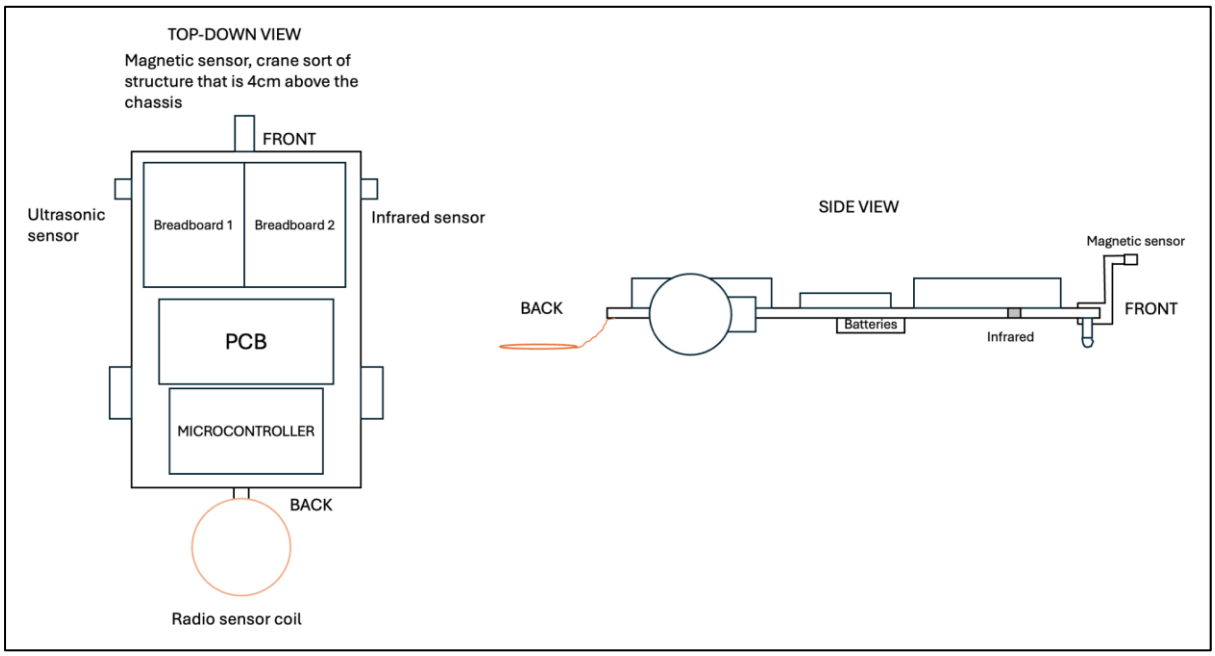


Fig 7.2: Rover Concept Design 1

The radio antenna coil will be at the back of the robot, ultrasonic and infrared sensors at the sides, and a magnetic field sensor elevated on a 3D printed Z-shaped mount for it. The mount is to elevate the magnetic field sensor, as the sensors team experimented and found out that the sensor not only needs to be parallel to the magnet to get an accurate reading, but be as close as possible, which means it needs to be within the ring on top of the duck's head. The movement concept would be for the robot to spin and use each sensor to gather the respective data to send to the app, before moving on to the next duck. The chassis concept design for it is as shown in Fig 7.3.

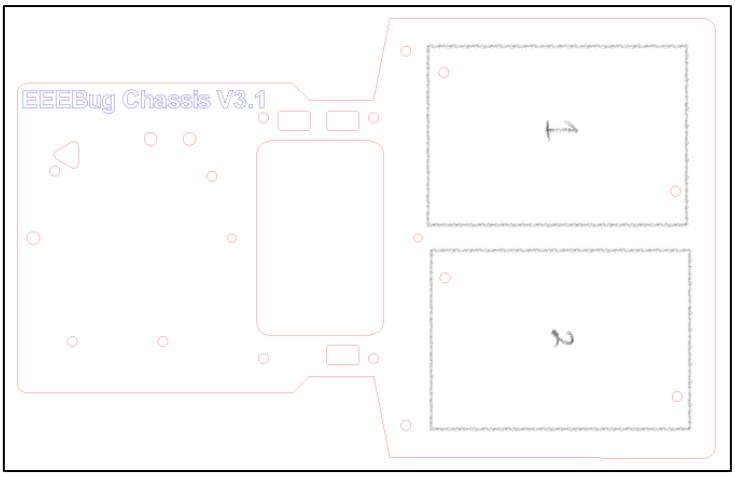


Fig 7.3: Rover Concept Design 1 Chassis

However, the team considered the main trade-off of this design, which is a wider front due to the breadboards being placed side by side. This would result in less manoeuvrability, especially under the bridge. It might also result in more weight due to a larger chassis.

The second design concept in Fig 7.4 is very similar to the first, with the main difference being the placement of the breadboards, which determine the width of our robot, which has been mentioned above, is restricted by the playfield. Rather than being placed side by side, the second breadboard is placed below the rover chassis.

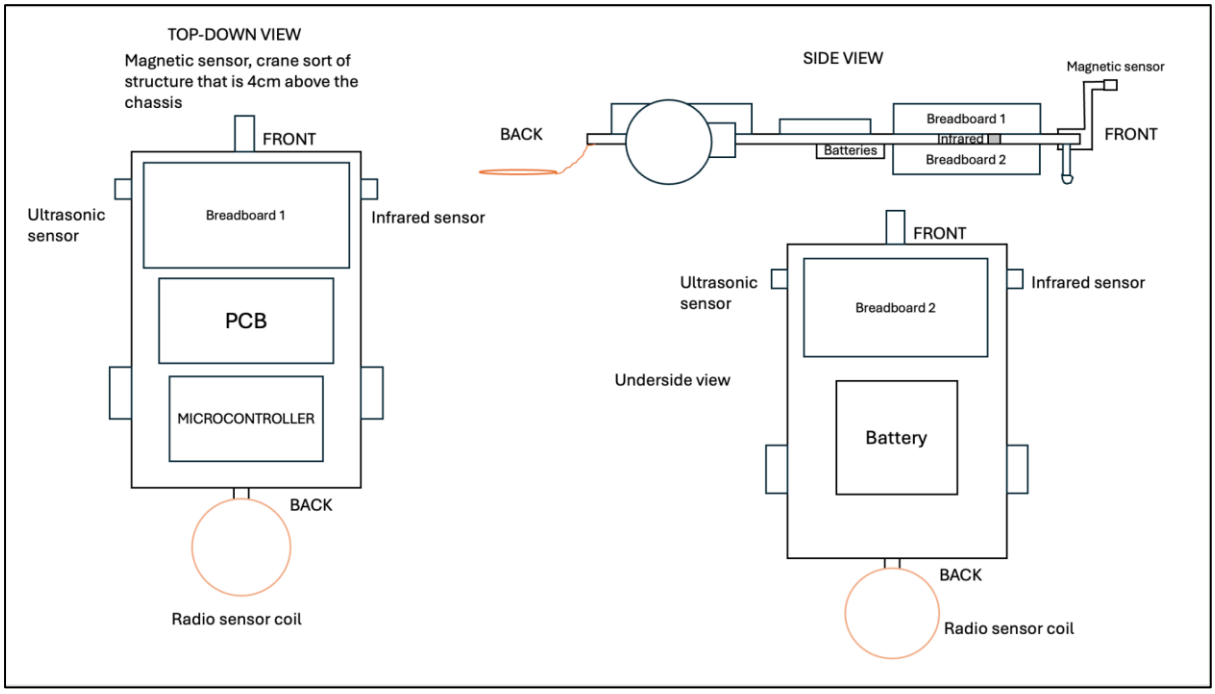


Fig 7.4: Rover Concept Design 2

Fig 7.5 shows the chassis design for the second concept.

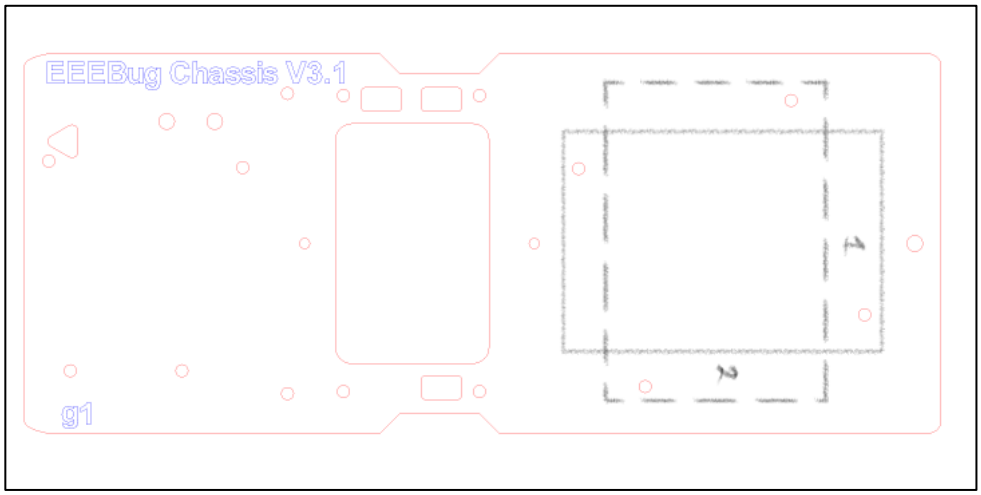


Fig 7.5: Rover Concept Design 2 Chassis

Although this design is smaller in width to keep within the size restrictions, it is also taller so as to prevent the bottom breadboard from dragging on the floor. This would mean larger wheels of 40mm radius are required. Similar mounts are required for the magnetic field sensor and radio antenna coil as mentioned above, but mounts are now also required for the ultrasonic and infrared sensors. This is because the chassis is now higher and will not be aligned to the body of the duck where the respective signal strengths are strongest. Hence to ensure accuracy of readings, mounts are required to lower the positions of the ultrasonic and infrared sensors.

Due to the above-mentioned reasons and rationale, the team decided to proceed with the second design. However, we decided to also include mounts to attach the radio antenna coil at the back to prevent dragging of the coil. Fig 7.6 shows the final updated rover design, while Fig 7.7 shows the final chassis design, which includes all holes for the screws.

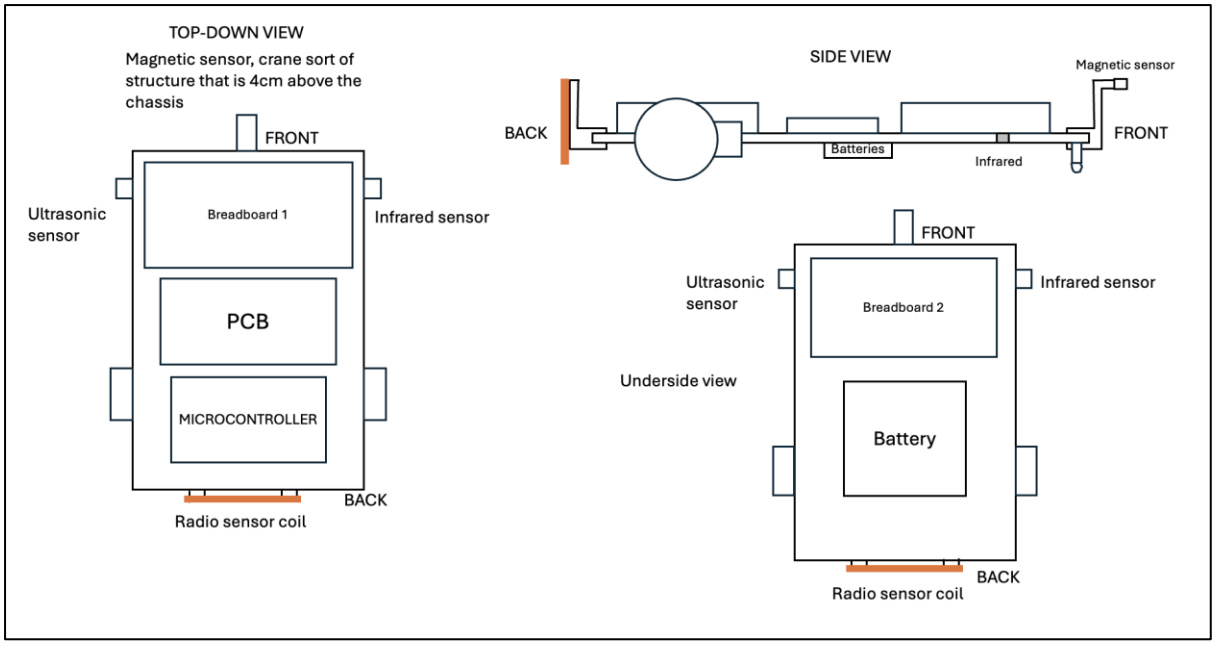


Fig 7.6: Rover Final Design

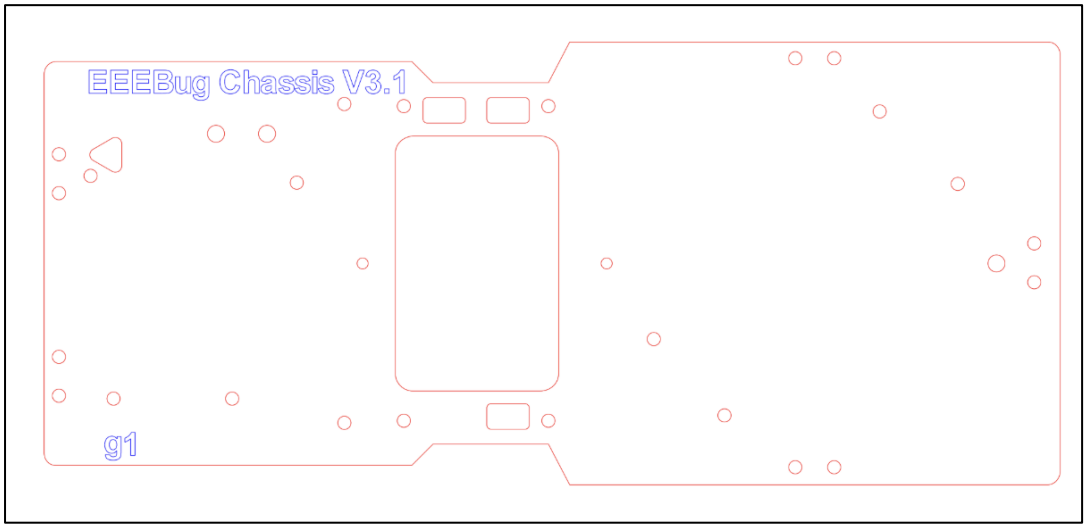


Fig 7.7: Rover Final Design Chassis

The team also designed the respective mounts as shown below for each sensor in Fig 7.8, 7.9 and 7.10.

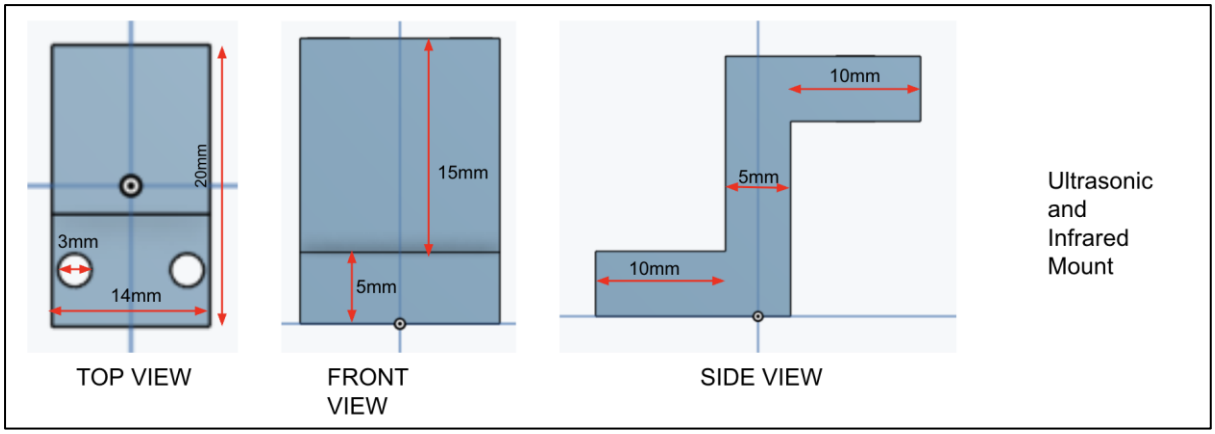


Fig 7.8: Ultrasonic and Infrared Sensor Mount Design

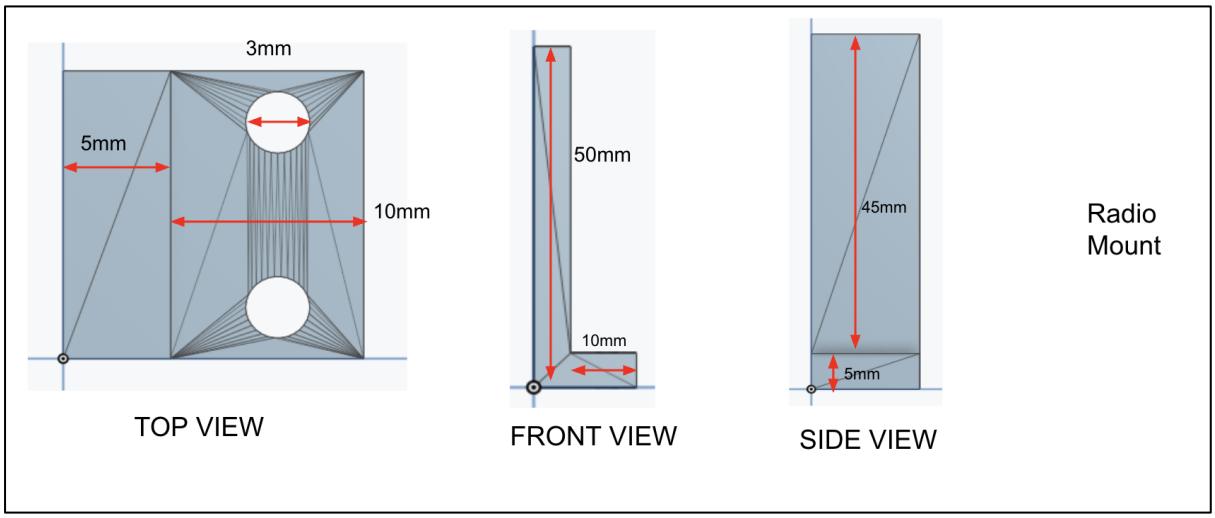


Fig 7.9: Radio Sensor Mount Design

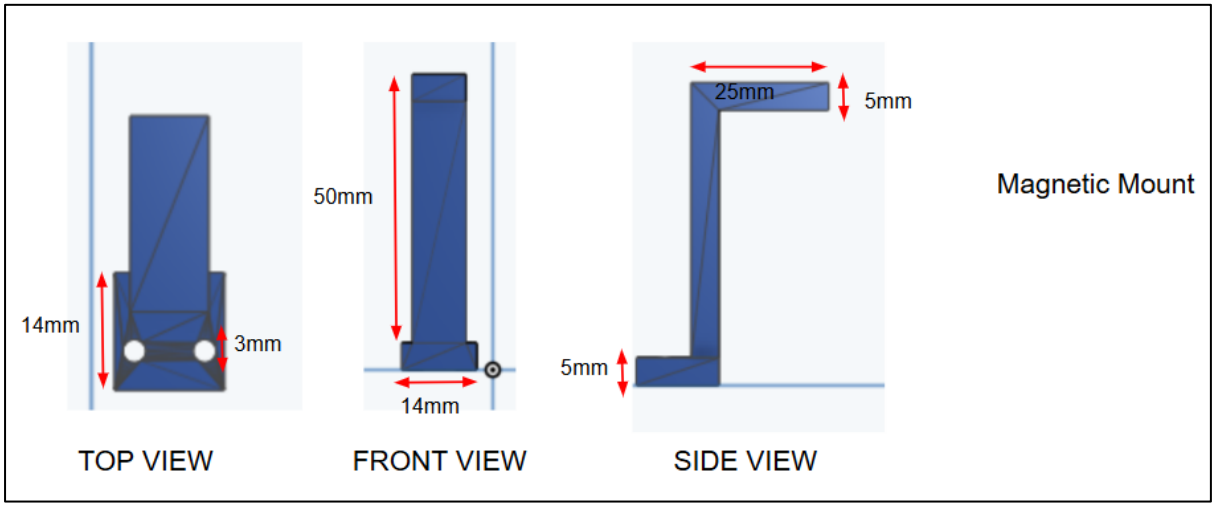


Fig 7.10: Magnetic Field Sensor Mount Design

Next, we designed the new, larger wheels, as shown in the Computer-Aided Design (CAD) renderings below in Fig 7.11 and 7.12.

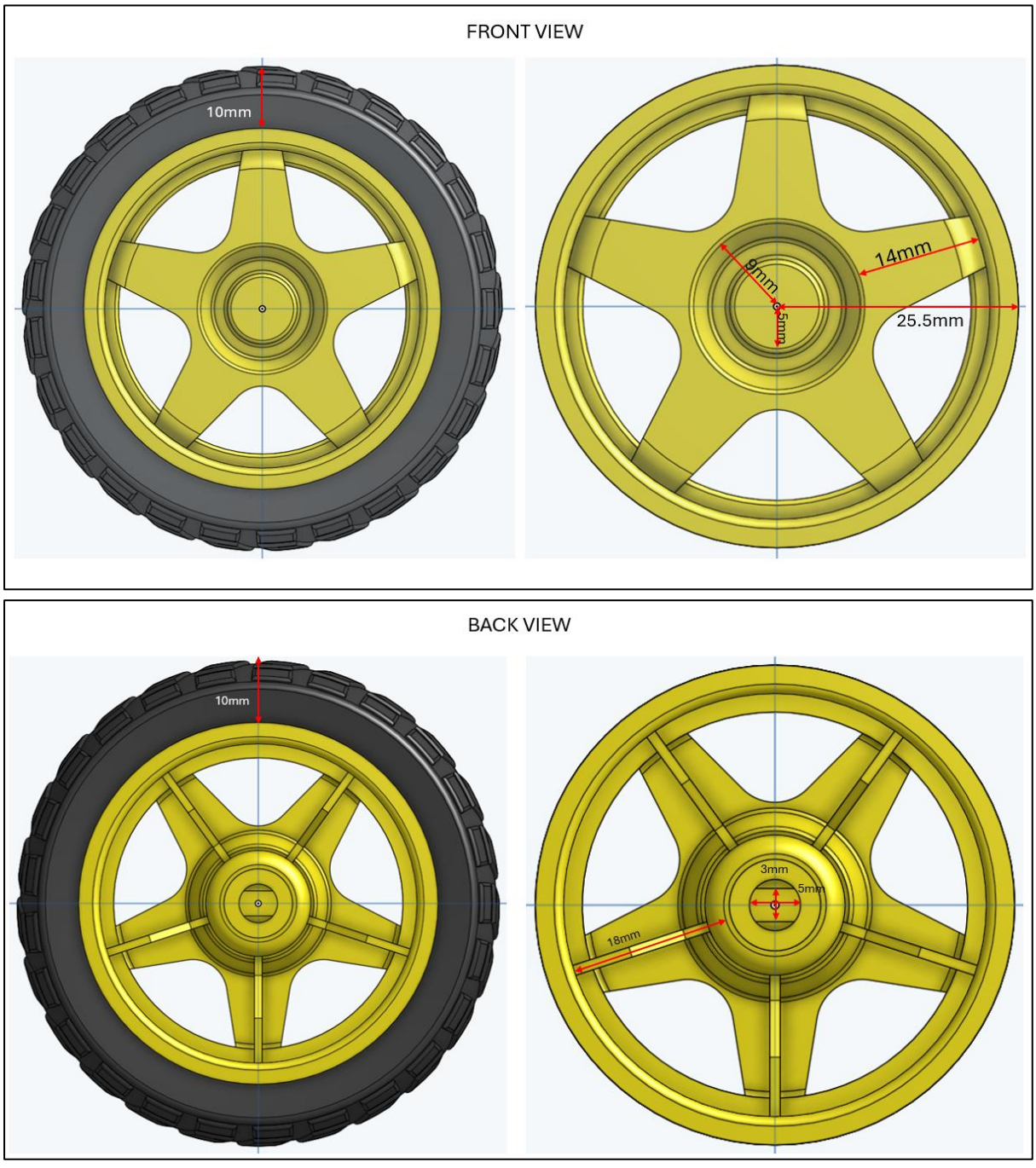


Fig 7.11: CAD Rendering of Wheel Design (1)

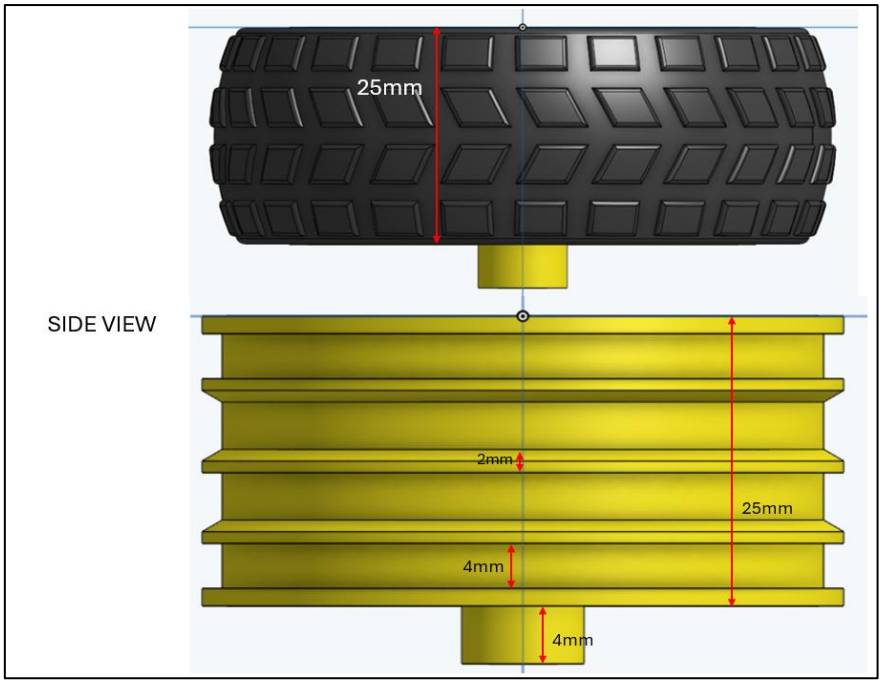


Fig 7.12: CAD Rendering of Wheel Design (2)

The wheel was designed this way to ensure and improve the vehicle's performance and stability.

A smaller diameter, narrower profile, and lighter weight characterised the original EEEBug wheels. Even though this led to a lower overall mass, it still brought up limitations in performance in terms of ground clearance and traction. Hence, the radius of the wheels was increased, and a reinforced five-spoke hub for the wheel, with a textured tread for enhanced grip.

The increased width and tread pattern improve the grip on the surface, reducing slippage and allowing the rover to be controlled with higher precision. The larger diameter accommodates the second breadboard at the bottom and will enable the rover to better clear obstacles. The five-spoke hub and thicker rim improve the structural integrity, making it more robust.

7.2 Technical Solutions and Build

Fig 7.13 shows the final build of the rover, with all sensors and circuits integrated onto it.

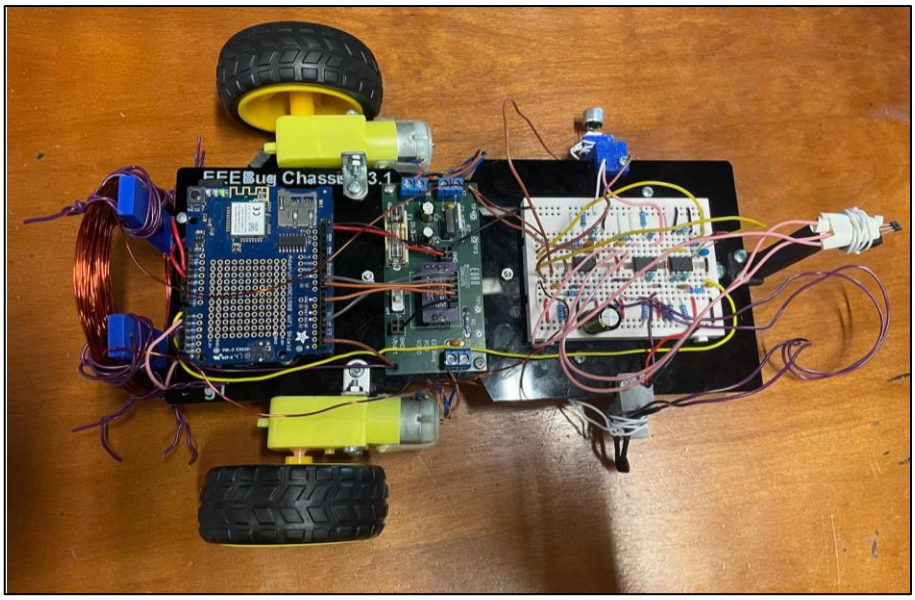


Fig 7.13: Final Rover Build

Next, we perform testing and measurements to ensure that the final rover meets all task requirements and specifications. Firstly, we tested and managed to ensure that all four sensors and subsystems work consistently and can relay accurate information to the user via the app. On the mechanical side, the final rover has a weight of about 600g. It is also 252mm long, 190mm wide and 80mm tall, which all fall under the size restrictions imposed on the rover. The app was also able to smoothly control the movement of the rover.

8 Remote Control UI

8.1 Design Overview

Fig 8.1, 8.2 and 8.3 shows the flowcharts that give an overview of the code developed for each subsystem to process the data received by each sensor subsystem.

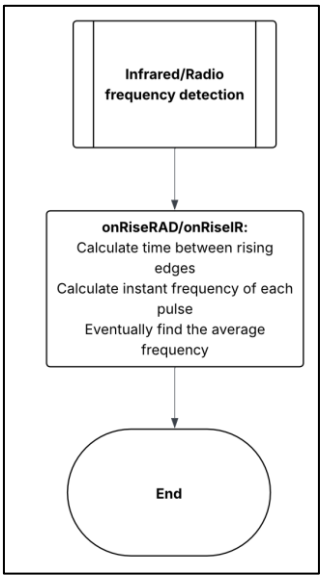


Fig 8.1: Flowchart Subroutines for Infrared/Radio Signal Detection

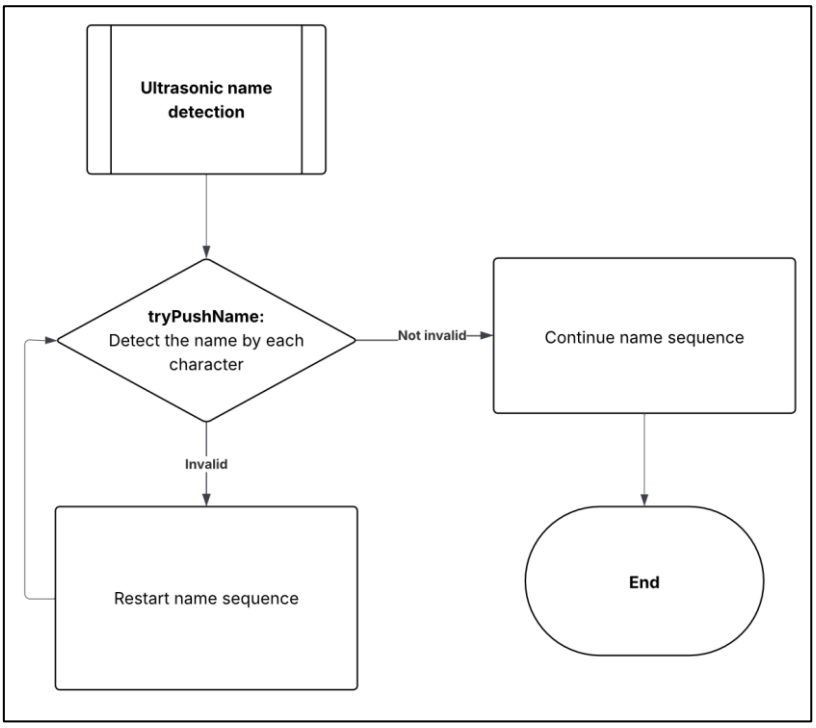


Fig 8.2: Flowchart Subroutines for Ultrasonic Signal Detection

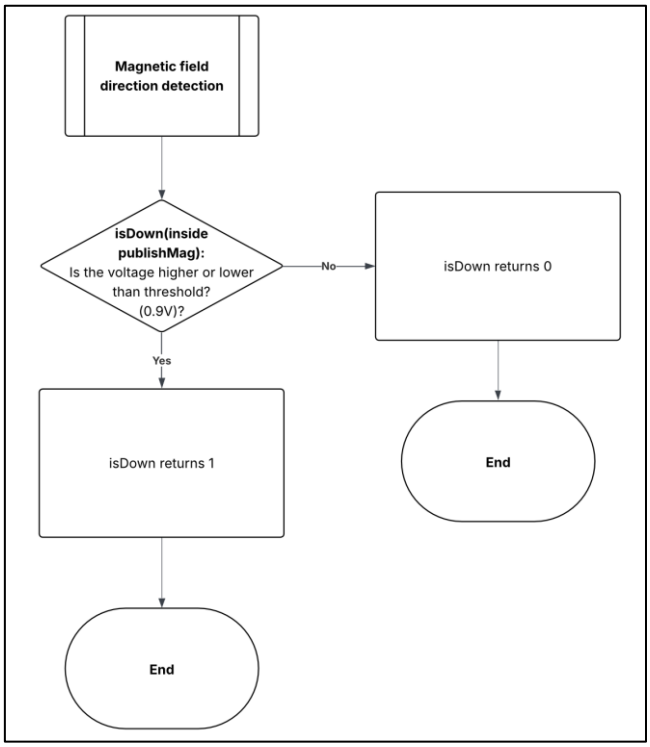


Fig 8.3: Flowchart Subroutines for Magnetic Field Direction Detection

The outputs from the respective subsystem subprocesses are then compared to existing values provided to them to categorise and determine the species of the duck.

Fig 8.4 shows a flowchart which encompasses all subroutines above, giving an overview of the communication workflow of the WiFi-enabled rover with multiple sensors and MQTT-based messaging.

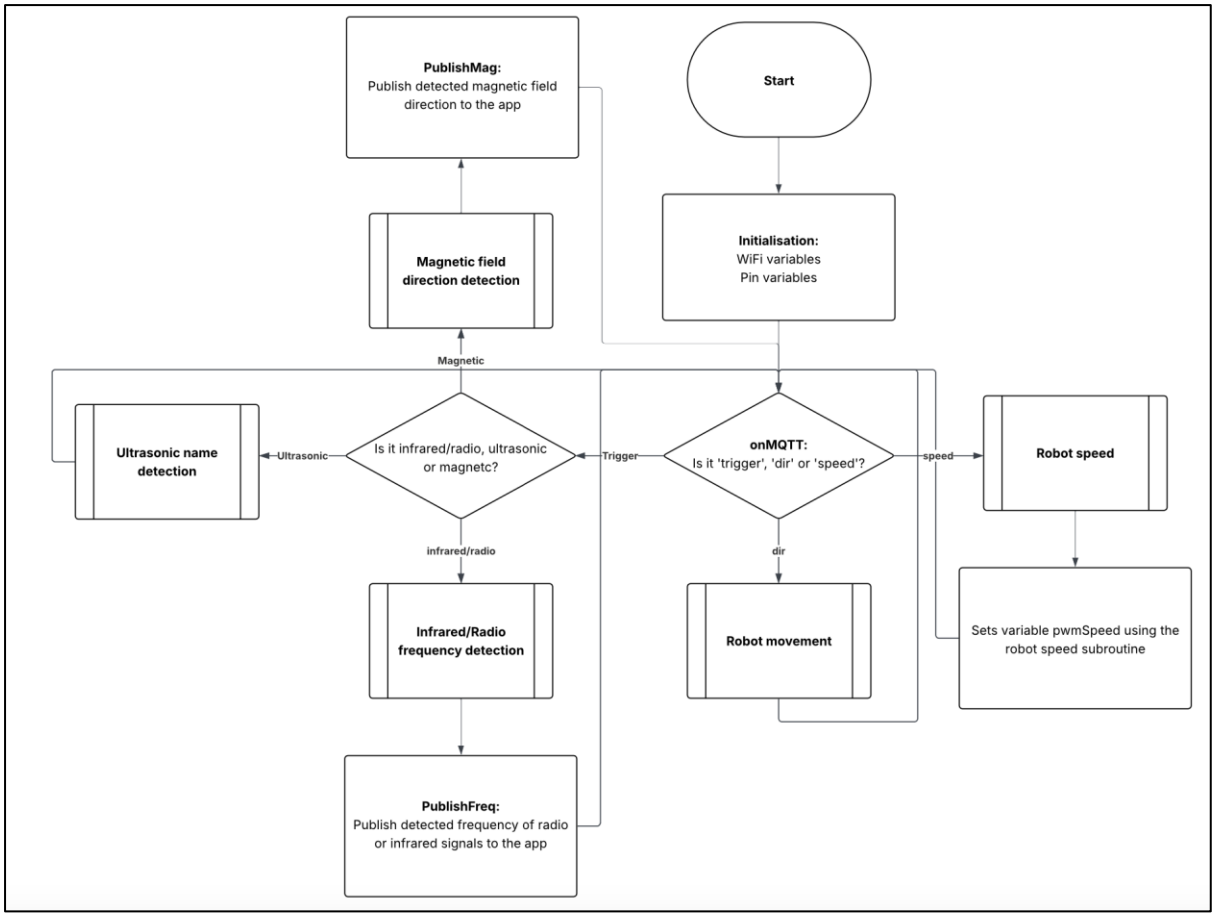


Fig 8.4: Flowchart for Overall Communication Workflow

The rover receives control commands remotely, interacts with its sensors, and transmits the sensor data back to the central controller via WiFi and MQTT.

When booted up, the rover connects to a preconfigured WiFi and is handled in the ensureWifi() function. If no connection is made in 15 seconds, the system will reset the Wifi connection and retry the connection, so it establishes a network link that allows us to communicate with the MQTT broker. When the WiFi is established, the rover initialises the MQTT client, which allows real-time bidirectional communication with our remote controller.

Incoming MQTT messages are processed via the onMqtt() function, which is in the JSON format, and will contain the trigger or direction commands. The trigger key instructs the rover to begin collecting data from a sensor. Once sufficient samples are acquired via interrupt routines, the results are averaged and published using MQTT. The “dir” key specifies a direction string, and the functions execDir(dirStr) and echoMotion(dirStr) will allow the rover to be remotely controlled and respond to environmental sensing requests.

The rover uses MQTT to publish the results of its operations to a public topic on the screen of our remote control within the app. The sensory frequency readings are averaged and sent using

the publishFreq() function, and there are heartbeat messages sent every 30 seconds to signal that the rover is alive and connected.

The main loop is made to sustain the operation of the system, the connectivity, and the data handling. The WiFi and MQTT monitoring ensure the communication channels remain active. It is also responsible for the sensor data decoding; The microcontroller then directly parses UART code through Pin0. This function implements a simple state machine for identifying three-letter names such as "Bip" and "Eva". It determines whether the character format is consistent in sequence: the first character is an uppercase letter, and the next two are lowercase letters. Once a successful match is found, the name is constructed into a JSON format message and published to the remote-control end. Every 30 seconds, it also sends out a keep-alive signal to check its operational status.

There is interrupt-driven sensor sampling for precise data collection; these detect rising edges from the respective sensors, allowing the rover to count frequency changes efficiently and without delay.

Moving on to the UI of the app, the team came up with the following concept design in Fig 8.5.

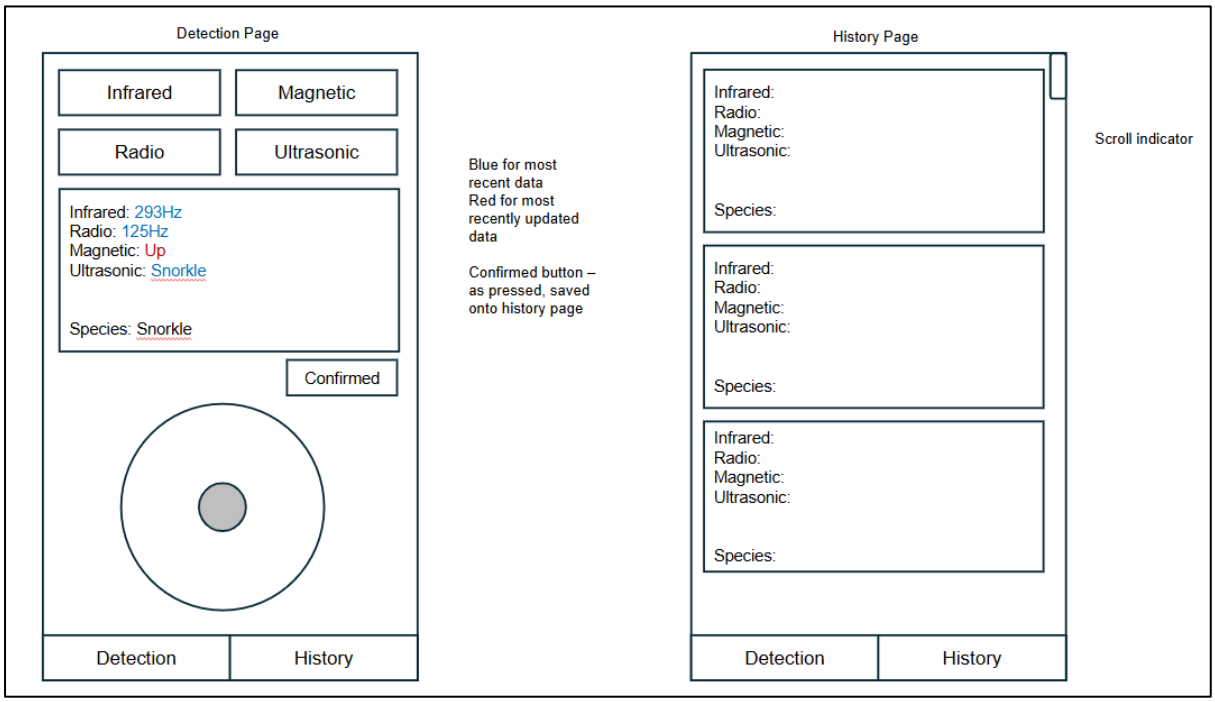


Fig 8.5: App UI Concept Design

As part of the final system integration, a refined app interface was created to enhance user interaction and experience. It also streamlines the data acquisition and allows us to have better control over the robot. Users press a designated function button to initiate a data request from a particular sensor, which is displayed on the screen. The screen always displays the most

recent reading from each sensor, and when new data is received, the field temporarily turns red, giving us confirmation that the data has been updated. Based on the user data, the system compares the input with known values provided to it beforehand and displays the identified duck species on the screen. Once a species has been correctly identified, the user can press the "Confirmed" button, which saves all the current data fields to the app's history page. The history page can be toggled via a button at the bottom of the interface and displays a scrollable list of all the previously identified ducks.

8.2 Technical Solutions and Build

Fig 8.6 shows the most initial version of the app UI. In earlier testing phases, this app UI used a much more rigid control scheme, and speed was fixed using three fixed buttons, requiring users to first set a speed and then press a separate direction button.

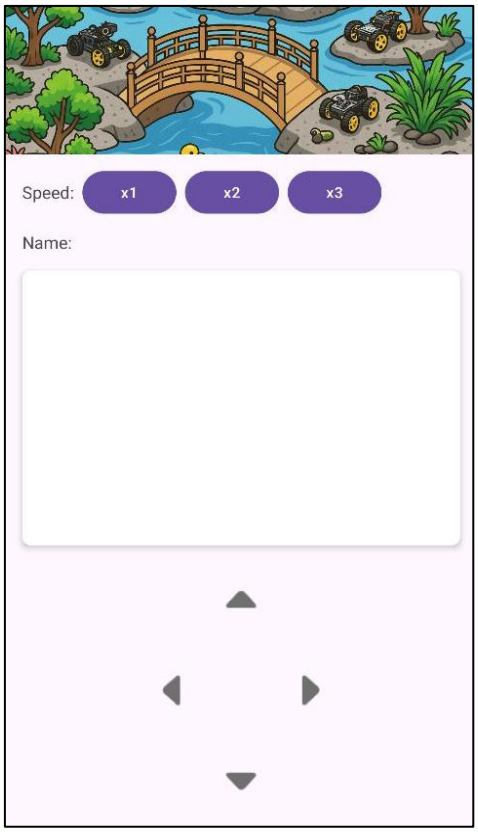


Fig 8.6: Initial App UI

During testing, this was found to be clunky and unintuitive, especially when quick adjustments needed to be made. Furthermore, the speed selected was easily forgotten, and it was cumbersome to switch between speeds and directions during navigation. Hence, the button-based movement was replaced with a joystick control, allowing analogue input for both speed

and direction in a single gesture, improving the fluidity of the movement. How much the joystick is dragged from its original position determines the speed of the robot. Hence the second version of the app UI is as in Fig 8.7.

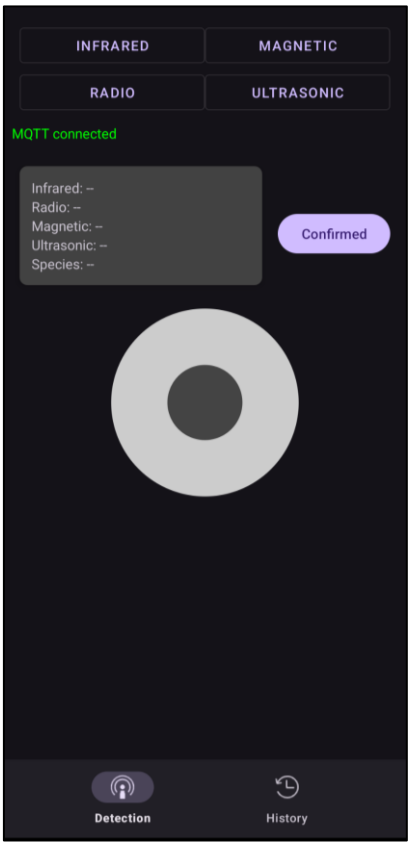


Fig 8.7: Final App UI

Overall, it offered improved usability and real-time feedback, as well as allowing streamlining of data management with the confirmation button and history log. There was also the more intuitive navigation with the joystick control, which supports the efficient identification and tracking of the duck species.

9 Integrated Testing and Discussion

9.1 Power Testing

As seen from all circuits above, the power supply required is from -3.3V to 3.3V. Although this is possible in testing by using the benchtop power supply, a voltage regulator is required so that the rover's microcontroller can provide the power supply range required. The sensors team found the ADM8660[1] voltage regulator, which would help provide the negative voltages. This voltage regulator has a voltage input range of 1.5V to 7V, which means the 3.3V required would be viable. It also has a power efficiency of 94% and a voltage conversion efficiency of 99% under no load, which allows the microcontroller to provide the entire power supply range required by the circuits.

Based on the data as stated above, the team has the following circuit design in Fig 9.1 for the voltage regulator. For the voltage regulator to work correctly we used the typical applications circuit from the datasheet.

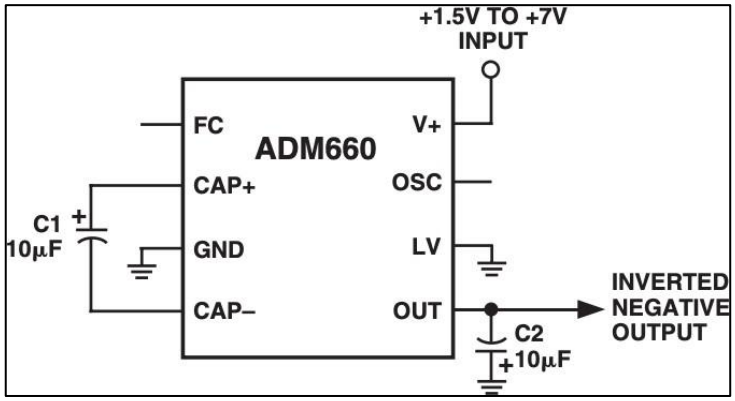


Fig 9.1: Circuit Diagram for Voltage Regulator

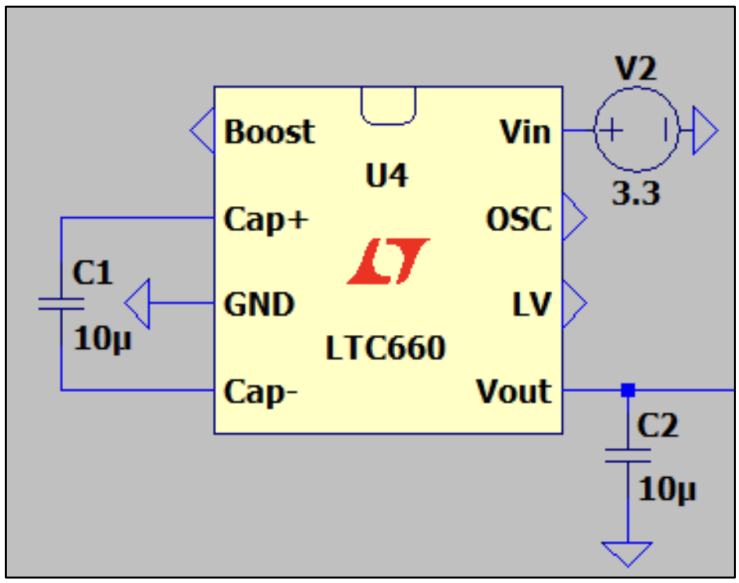


Fig 9.2: LTspice Diagram for Voltage Regulator

As we were unable to find the ADM8660 voltage regulator on LTspice, we decided to model it using a similar one. Do note that two of the pins on the chosen model are incorrect, the Boost pin and the OSC pin, as they should be FC and SD instead. However, the connections for both are to ground, as shown above.

Based on the circuit design shown above in Fig 9.2, the circuit for the voltage regulator in Fig 9.3 was built.

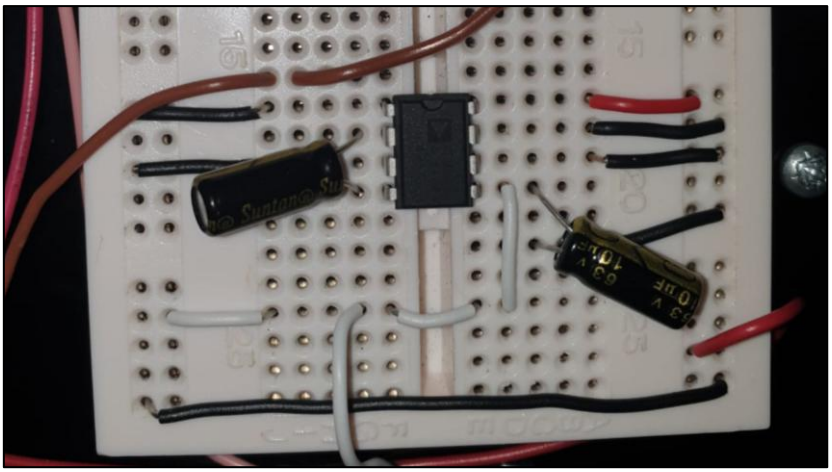


Fig 9.3: Voltage Regulator Circuit

Testing was also done to ensure that the voltage regulator consistently works as required.

9.2 Integration and Testing

During integration and testing, all the circuits were put together. The sensors team decided to also purchase a new op-amp for the circuits. The original op-amp, TL072CP, was not sufficient in achieving the task requirements. This op-amp was not rail-to-rail, causing the output to be saturated at around 2V, even though the power supply was 3.3V. It also had a low gain-bandwidth product of 5MHz. This was an issue as when the team tried to have a gain of 100 for the radio signal which has a carrier frequency of 89kHz, the actual gain was much lower due to the gain-bandwidth product. To obtain sufficient amplitude, two gain stages were required instead, doubling the complexity of the circuits.

Hence, the team researched and found the MCP6022-E/P op-amp. This op-amp is rail-to-rail, which is more appropriate for low-voltage applications like the above circuits. It also has a higher gain-bandwidth product of 10MHz, removing the issue faced above as one gain stage was sufficient to obtain the amplitude required. Overall, it is much better to achieve the task requirements in this project.

In the circuits shown above, we were unable to find the MCP6022-E/P in LTspice and hence decided to use one with features such as the gain-bandwidth product to obtain similar simulation results.

Fig 9.3 shows our final circuits, while Fig 9.4 shows our final rover.

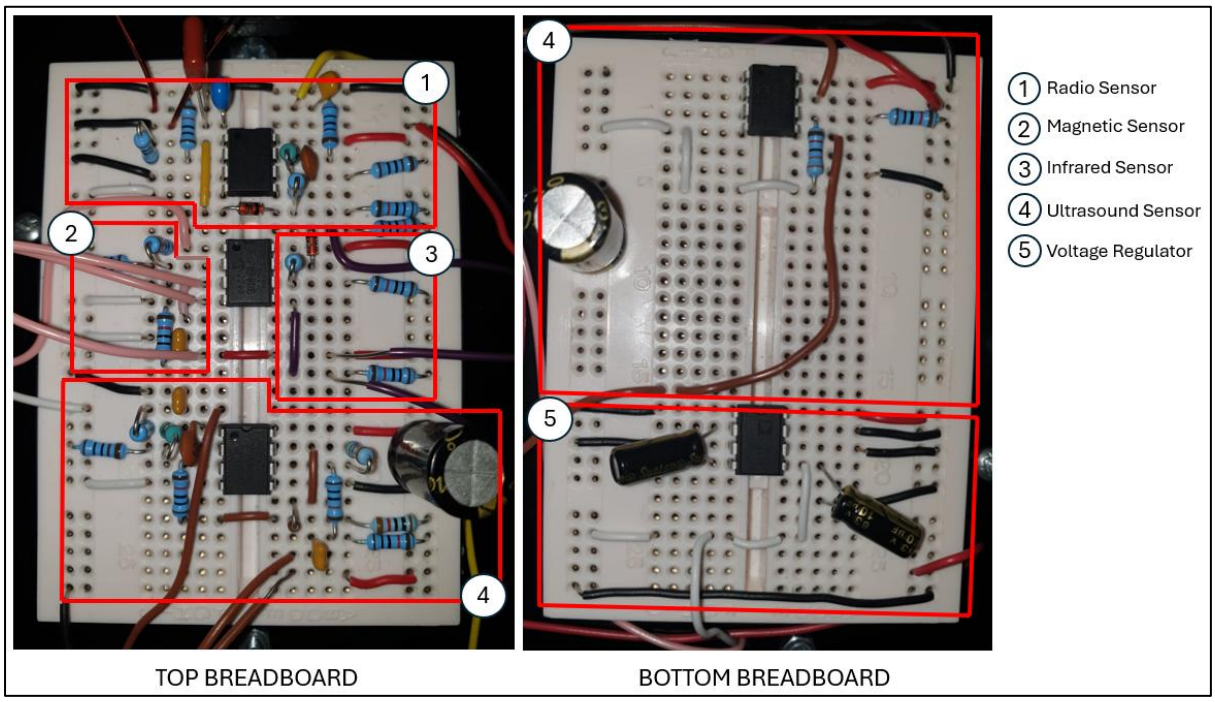


Fig 9.4: Final Circuit

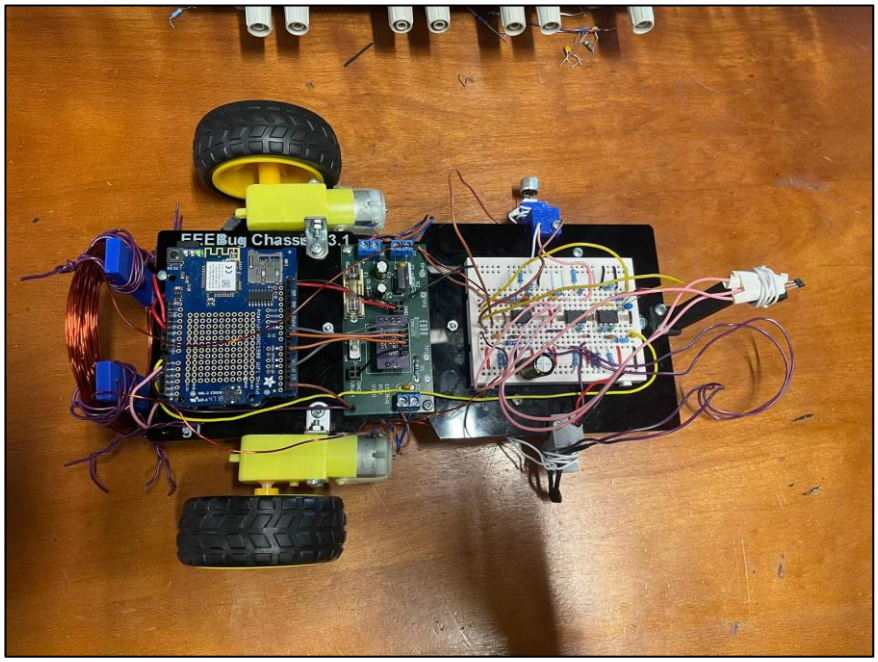


Fig 9.5: Final Rover

9.3 Discussion

Overall, all our sensor circuits work as expected. For the radio and infrared, we were able to obtain the correct frequency according to the duck setting. For our ultrasound circuit, we were able to obtain the personalised name of our duck: “Bip”, as well as the random names that made sense. We were also able to obtain the correct direction for the magnetic field with our magnetic sensor circuit, and the circuit and analysis results on our app responded well when we changed the orientation of the field.

During the process of working on the project, we encountered some problems. During component sourcing, we misinterpreted some of the products’ specifications, which led us to purchase the wrong product. For example, we initially purchased an ultrasonic transmitter instead of a receiver. While this led to some additional cost and a slight delay, it emphasized the importance of thoroughly verifying datasheets and part numbers prior to purchase. This experience helped us improve our component selection process moving forward. Other than that, we also faced the issue of components not being securely attached onto the breadboard. This caused many of our tests to fail as op-amps were occasionally coming loose, which resulted in time inefficiencies. A possible improvement would be to use a protoboard instead where we can solder components onto the board so that our circuit becomes more robust.

9.4 Budget

Fig 9.5 below shows a record of the team’s expenses and budget.

Component	**Quantity**	**Price/Unit (£)**	**Total Cost (£)**	**Budget Left (£)**
CA3140E Single Op-amp, 4.5MHz DIP8	3	0.5	1.5	11.32
Diode Zener 1.3W BZX85 3.3V	2	0.42	0.84	
Diode IN5408 3A 1KV	3	0.21	0.63	
Diode IN916 75MA 100V MT 4NS	5	0.2	1	
Allegro Microsystems Through Hole Hall Effect Sensor, SIP, 3-Pin	2	1.3	2.6	
RS PRO Ultrasonic Smooth Barrel Proximity Sensor, 300 mm Detection, 20 V	1	13.25	13.25	
TL072CP General Usage Dual Op-amp, DIP8	5	0.52	2.6	
400SR100 Receiver, Ultrasonic, 40KHZ, 10MM	1	5.68	5.68	
MCP6022-E/P Operational Amplifier, Dual, 2 Channels, 10 MHz, 7 V/µs, 2.5V to 5.5V, DIP, 8 Pins	5	1.56	7.8	
BPW82 Photo Diode, Non Amplified, 950 nm, Radial Leaded,	3	0.88	2.64	

65 °				
Analog Devices ADM8660ANZ Charge Pump, Regulator, -7 → -1.5 V 8-Pin, PDIP	2	5.07	10.14	

Fig 9.5: Expense and Budget Spreadsheet

Most of the initial team expenses were for components available in the EEE Stores Catalogue, allowing us to begin trials and testing immediately without waiting for deliveries. Meanwhile, the teams conducted research and ordered the additional components and sensors needed for the rover. This approach enabled efficient development and testing of each rover subsystem, minimizing delays.

10 Conclusion

In this project, we successfully designed, built, and tested a rover featuring an app-based user interface and a mechanical structure optimized for signal detection. The sensors team also implemented and integrated circuit systems capable of processing signals from the sensors, enabling the microcontroller to process the outputs accordingly. Furthermore, we developed software which enabled seamless communication with the rover for both control and data transmission, as well as process the information received to determine the ducks' name and species.

Overall, this project provided us with the opportunity to develop and apply a wide range of both technical and soft skills.

We learned how to conduct targeted research to identify components that met our design requirements, ensuring that subsystem outputs could be accurately read and processed by the microcontroller. For instance, we had to research about op-amps to determine if we could use a single gain stage to provide enough amplification for our inputs.

In addition, we learnt more about circuit design, exploring the roles and functions of various components. Our circuits primarily involved amplifiers, demodulators and comparators, allowing us to consolidate and apply concepts learnt in the course.

However, the scope of the project extended beyond the taught curriculum, such as working with magnetic sensors, which required independent research and self-learning.

Finally, the project also enabled us to apply our software knowledge to handle both digital and analogue data, such as decoding UART signals from ultrasonic sensors to identify the duck's name. This integration of hardware and software enriched our overall learning experience and helped us bridge theory with practical implementation.

This project also helped us develop valuable soft skills that are essential in a professional environment. We understood the importance of teamwork, recognising that each member brings a unique skill set. Only by combining our strengths were we able to tackle the diverse

challenges presented throughout the project—an experience that reflects the collaborative nature of real-world engineering work.

Effective communication was another key takeaway. To keep the project on track, we had to ensure that everyone was aligned and informed. For example, it was crucial for the sensors team to inform the mechanical and UI team of their findings so that these findings could be taken into consideration for the rover design.

Lastly, we improved our time management by delegating tasks, setting clear milestones, and adhering to a project schedule. This allowed us to complete the project within the given timeframe while maintaining quality across all subsystems.

11 References

BPW82 Photodiode/Infrared Sensor Datasheet:
https://www.farnell.com/datasheets/2240236.pdf

MCP6022-E/P Op-Amp Datasheet:
https://4donline.ihs.com/images/VipMasterIC/IC/MCHP/MCHPS02737/MCHPS02737-1.pdf?hkey=6D3A4C79FDBF58556ACFDE234799DDF0

400SR100 Ultrasonic Receiver Datasheet: https://www.farnell.com/datasheets/4413607.pdf

Allegro Microsystems Magnetic Hall Effect Sensor:
https://www.allegromicro.com/~/media/files/datasheets/a1324-5-6-datasheet.ashx

Analog Devices ADM8660ANZ Charge Pump, Regulator, -7 → -1.5 V 8-Pin, PDIP:
https://docs.rs-online.com/79cf/0900766b80e46aa3.pdf